

The Mézard–Parisi formula for diluted spin glasses

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Abstract

We prove the Mézard–Parisi hierarchical cavity formula for diluted even-arity Ising models in the Panchenko–Talagrand class. This resolves the variational equality conjecture for that class: the limiting pressure equals the infimum of the trial functional over all finite hierarchy depths and trial laws. Only first moments of the interaction and external field are required.

1 Introduction

A diluted spin glass has a finite mean number of interactions per spin. Its local environment remains random as the system grows, and the cavity method describes that environment through a distribution of local fields. When many competing states contribute to the partition function, the proposed order parameter is a hierarchy of distributions of these fields. The thermodynamic question is whether this hierarchy determines the limiting expected log partition function per spin. We call this quantity *pressure*. At inverse temperature β , the usual physical free energy per spin is its negative divided by β .

The Viana–Bray model is a classical diluted two-spin model [18]. Mézard and Parisi [12] developed the hierarchical cavity picture for the fixed-connectivity Bethe-lattice spin glass, with an explicit one-step replica-symmetry-breaking functional. Franz and Leone [9] adapted Guerra’s interpolation method [11] to diluted systems and obtained replica-symmetric and one-step pressure bounds for even-arity models. Panchenko and Talagrand [16] then proved the arbitrary finite-level hierarchical upper bounds considered here.

We prove that the infimum of these bounds is the limiting pressure. This resolves the equality conjecture stated immediately after Theorem 4 of Panchenko–Talagrand [16], for their Poisson, even-arity Ising model class with factorized interactions and the specified positivity condition. Only first moments of the interaction and external field are required. The result includes the Viana–Bray model, diluted even-spin models with symmetric couplings, and soft random even- K SAT at every positive temperature. The exact model and trial functional are defined in [Section 2](#); the named models and their expected ground-state limits are treated in [Section 10](#).

An exact variational description and reduction to a prescribed trial class are distinct problems. The cavity variational principle of Aizenman, Sims and Starr [2] for the Sherrington–Kirkpatrick model has a Viana–Bray counterpart in De Sanctis’s formulation over random multioverlap structures [7]. Panchenko’s spin-distribution approach [14, Theorems 1–2] gives an exact variational formula over invariant exchangeable spin distributions. Under their positivity condition, Coja-Oghlan and Perkins [5] obtain an invariant-kernel variational formula for random regular factor graphs. These results use different descriptions of the state from the finite hierarchy of message laws optimized here. The additional task is to recover the specific hierarchical cavity functional without losing the pressure.

Panchenko [15, Theorem 2] established a hierarchical single-site representation for finite-RSB asymptotic Gibbs measures of suitably modified models; approximation of general states by that class remains a separate step. Biswas, Chen and Sen [4, Theorem 1.1 and Proposition 1.4] obtain exact replica-symmetric pressure formulas in high-temperature and critical or subcritical regimes for compact real spin spaces under moment and regularity assumptions. For the zero-field Ising spin glass with Rademacher couplings on random regular graphs, Concetti [6, Theorem 1.3] gives a full-replica-symmetry-breaking pressure bound no larger than each optimized finite-level bound; equality with that functional is conjectural in that work. The result here identifies the finite-depth Panchenko–Talagrand infimum at arbitrary density in its stated interaction class.

The main obstacle is the distribution of spins seen across several replicas, that is, several samples from the same Gibbs system. The probability of each replica-spin pattern depends on products of every subset of those spins. The spatial average of such a product is a *multioverlap*; pair overlaps alone do not determine all these patterns. This is the information isolated by the spin-distribution approach [14]. To compare a cavity insertion with independent hierarchical messages, we must control all finite multioverlaps before redrawing the message roots independently.

We work with finite auxiliary hierarchies before taking the thermodynamic limit. Their branching is part of the construction; no ultrametric organization of the physical Gibbs states is assumed. Bounded perturbations yield identities for prescribed replica branching patterns. Delaying each branching depth by one level permits an induction that concentrates multioverlaps *on average over branching depths*. The cavity insertion itself has an expansion averaging over finite trees, so this averaged control is sufficient. The system size tends to infinity at each fixed depth; only then does the depth increase. The proof strategy following Theorem 2.1 explains how the finite-tree calculus, concentration and independent-message comparison fit together.

2 The model and the variational principle

Fix an even integer $p \geq 2$ and $\alpha > 0$. Let θ be a random real function on $\{-1, 1\}^p$ and let h be a real random variable. Write $B(\theta) = \max_s |\theta(s)|$ and assume

$$\mathbb{E}B(\theta) + \mathbb{E}|h| < \infty. \quad (2.1)$$

For independent $M_N \sim \text{Poisson}(\alpha N)$, independent copies θ_k of θ , independent copies h_i of h , and independent uniform indices $i_{\ell,k} \in [N] = \{1, \dots, N\}$, set

$$\begin{aligned} -H_N(\sigma) &= \sum_{k=1}^{M_N} \theta_k(\sigma_{i_{1,k}}, \dots, \sigma_{i_{p,k}}) + \sum_{i=1}^N h_i \sigma_i, \\ Z_N &= \sum_{\sigma \in \{-1, 1\}^N} e^{-H_N(\sigma)}, \quad F_N = \frac{1}{N} \mathbb{E} \log Z_N. \end{aligned} \quad (2.2)$$

All families in this definition are mutually independent. In particular, indices are sampled with replacement. The model parameters and disorder laws are fixed as N varies.

The interaction hypotheses are the existence of a representation

$$e^{\theta(s_1, \dots, s_p)} = a \left(1 + b \prod_{\ell=1}^p f_{\ell}(s_{\ell}) \right), \quad |b \prod_{\ell=1}^p f_{\ell}(s_{\ell})| < 1 \quad \text{a.s.}, \quad (2.3)$$

for every spin tuple, where $a > 0$ and b are real random variables, the $f_{\ell} : \{-1, 1\} \rightarrow \mathbb{R}$ are independent copies of one random function, b is independent of them, and

$$\mathbb{E}[(-b)^n] \geq 0 \quad (n \geq 1). \quad (2.4)$$

The expectations in (2.4) are interpreted as finite real expectations. This is automatic in the nondegenerate case: independence and the strict product bound force b and $\max_s |f_1(s)|$ to be essentially bounded, as shown in Proposition 4.1. If the product term vanishes identically, the interaction is spin independent and one may take $b = 0$. The multiplier a may depend on all the other variables in the representation; no separate integrability of $\log a$ is assumed.

2.1 The hierarchical cavity functional

We give the power-mean form of the Panchenko–Talagrand functional [16, Section 5]. A message x specifies a probability distribution on one spin. The site term inserts a new spin with its incident interactions; the edge term corrects for interactions counted in that insertion.

For $x \in \mathbb{R}$ let $q_x(s) = e^{xs}/(2 \cosh x)$. Define

$$C_\theta(x_1, \dots, x_p) = \sum_{s \in \{-1, 1\}^p} e^{\theta(s)} \prod_{\ell=1}^p q_{x_\ell}(s_\ell), \quad (2.5)$$

$$U_\theta(x_1, \dots, x_{p-1}; \varepsilon) = \log \sum_{s \in \{-1, 1\}^{p-1}} e^{\theta(s_1, \dots, s_{p-1}; \varepsilon)} \prod_{\ell=1}^{p-1} q_{x_\ell}(s_\ell). \quad (2.6)$$

Both $|\log C_\theta|$ and $|U_\theta|$ are bounded by $B(\theta)$, uniformly in the messages.

Write $\mathcal{P}(E)$ for the Borel probability measures on E . Let $\mathcal{M}_1 = \mathcal{P}(\mathbb{R})$ and $\mathcal{M}_{j+1} = \mathcal{P}(\mathcal{M}_j)$, equipped with the topology of weak convergence and its Borel sigma-field. Fix $r \geq 1$, $\zeta \in \mathcal{M}_{r+1}$, and $0 < m_1 < \dots < m_r < 1$. For each message label ω independently, draw

$$\eta_0^\omega \sim \zeta, \quad \eta_{d+1}^\omega \mid \eta_d^\omega \sim \eta_d^\omega \quad (0 \leq d \leq r-2), \quad x^\omega \mid \eta_{r-1}^\omega \sim \eta_{r-1}^\omega. \quad (2.7)$$

At $r = 1$, this means that each label first draws a random probability measure η_0 with law ζ , then draws x with law η_0 . Higher levels repeat this sampling of measures. All these variables are independent of the physical disorder. Use different labels for $x_1, \dots, x_p$ and for x_ℓ^j , $j \geq 1$, $1 \leq \ell < p$. Take independent $k \sim \text{Poisson}(\alpha p)$, $\theta, \theta_1, \dots$, and h , and put

$$V_s = \frac{1}{2} \sum_{\varepsilon=\pm 1} \exp \left(h\varepsilon + \sum_{j=1}^k U_{\theta_j}(x_1^j, \dots, x_{p-1}^j; \varepsilon) \right), \quad (2.8)$$

$$V_e = C_\theta(x_1, \dots, x_p). \quad (2.9)$$

For $0 \leq d < r$, let $\mathcal{F}_d$ be generated by all the physical disorder in these expressions and all η_j^ω , $0 \leq j \leq d$. Set

$$T_r V = V, \quad T_d V = (\mathbb{E}[(T_{d+1} V)^{m_{d+1}} \mid \mathcal{F}_d])^{1/m_{d+1}} \quad (d = r-1, \dots, 0). \quad (2.10)$$

Only auxiliary messages are integrated in these conditional expectations. For example, at depth one, $T_0 V = (\mathbb{E}_x[V^{m_1} \mid \boldsymbol{\eta}_0])^{1/m_1}$. At depth two, each label draws $\eta_0 \sim \zeta$, then $\eta_1 \mid \eta_0 \sim \eta_0$, and then $x \mid \eta_1 \sim \eta_1$, and

$$T_0 V = \left(\mathbb{E}_{\boldsymbol{\eta}_1} \left[(\mathbb{E}_x[V^{m_2} \mid \boldsymbol{\eta}_1])^{m_1/m_2} \mid \boldsymbol{\eta}_0 \right] \right)^{1/m_1}.$$

Here bold symbols collect all message labels in V ; physical disorder is held fixed throughout, including in the two abbreviated displays. Define

$$\mathcal{B}_r(\zeta, \mathbf{m}) = \log 2 + \mathbb{E} \log T_0 V_s - \alpha(p-1) \mathbb{E} \log T_0 V_e, \quad \Phi_r = \inf_{\zeta, \mathbf{m}} \mathcal{B}_r(\zeta, \mathbf{m}). \quad (2.11)$$

At $r = 0$, the messages are independent with common law $\zeta \in \mathcal{P}(\mathbb{R})$ and the operators T_d are omitted; this defines $\mathcal{B}_0(\zeta)$ and Φ_0 .

The quantities in (2.11) are finite without any moment assumption on the trial messages. Indeed, the power-mean operators preserve pointwise multiplicative bounds, and

$$|\log T_0 V_s| \leq |h| + \sum_{j=1}^k B(\theta_j), \quad |\log T_0 V_e| \leq B(\theta). \quad (2.12)$$

Theorem 2.1. *Under (2.1)–(2.4), the thermodynamic limit exists and*

$$\lim_{N \rightarrow \infty} F_N = \inf_{r \geq 0} \Phi_r. \quad (2.13)$$

The right side is an infimum over finite hierarchies. The theorem does not assume that a trial law attains it or that a limiting infinite-depth Gibbs state has been constructed.

2.2 Proof strategy

The interpolation upper bound holds at every system size and every finite depth. We reproduce it in Section 4, including a normalization that avoids assuming integrability of the multiplier $\log a$. The lower bound has four steps.

1. *Select a cavity reservoir.* A reservoir is a common Gibbs system on N spins used to compare systems of sizes N and $N + 1$. For bounded interaction and field laws, add bounded Poisson perturbations whose number is sublinear in the system size. Their effect on the pressure vanishes. At each fixed hierarchy depth, select system sizes and deterministic perturbation parameters for which the difference between successive expected logarithms after the auxiliary power-mean recursion is at most an arbitrarily small error above the original pressure liminf. The same choices give concentration of every perturbation score (Section 5 and Lemma 6.2).
2. *Prescribe branching patterns.* An elementary calculus for finite probability trees expands insertions into signed sums over added replica paths. Independent marks select any prescribed branching pattern. Score concentration then gives identities for these signed sums (Section 3 and Theorem 6.5).
3. *Concentrate all multioverlaps.* Move each internal branching depth of a target tree one step later. The first fresh split at each new depth supplies the same small factor as the normalizing coefficient. A relative bound on the sum of absolute coefficients preserves control of errors after division. Conditional covariance then decays by induction on the number of replicas. A second identity concentrates the remaining conditional mean (Section 7).
4. *Recover independent cavity messages.* Encode a uniformly chosen reservoir site's entire finite hierarchy by successive conditional laws. Multioverlap concentration permits different site labels to be replaced by independent copies of this process in the cavity insertion. This yields an admissible trial value close to the selected increment (Section 8).

The estimates in Step 3 are averages over separated interior branching depths. The insertion expansion assigns vanishing probability to the exceptional trees, so these averages are sufficient for Step 4. No uniform concentration in the depth is needed. The resulting lower bound holds for every bounded interaction law invariant under permutations of its inputs, without factorization or positivity. Finally, normalization and estimates uniform over all trial laws remove boundedness and give Theorem 2.1 under the stated first moment assumption (Section 9).

3 Calculus on finite trees

A power-mean insertion admits two expansions. The logarithmic expansion will compare pressures; an expansion with one canceled replica factor will extract the perturbation identities. Both follow from one differentiation rule on trees. We also bound their coefficients and the probability of exceptional branching depths. Power-mean calculations for hierarchical trial laws appear in Mézard–Parisi [12, Appendix A] and Panchenko–Talagrand [16, Section 5]; the latter uses cascade weights introduced by Ruelle [17]. Here we derive the needed identities directly for finite sequences of arbitrary probability kernels, without assuming an infinite hierarchy.

3.1 Path kernels and insertions

Fix an integer $L \geq 1$ and numbers

$$0 = m_0 < m_1 < \cdots < m_{L-1} < m_L = 1.$$

A path has depths $0, \dots, L$. Its depth-zero variable is the root. For $1 \leq d \leq L$, the probability kernel K_d draws the next variable given the whole prefix through depth $d-1$; $\mathcal{F}_d$ is the prefix sigma-field. The root distribution is fixed throughout all insertions. In an r -level application, $L = r+1$: depths $1, \dots, r$ carry auxiliary variables, and depth L carries a Gibbs spin configuration. For the trial law (2.7) with $r \geq 1$, η_d is the depth- d variable for $0 \leq d < r$, and x is the depth- r variable.

A finite tree $\mathcal{T}$ consists of a common root and finitely many labeled paths, each ending at depth L . Every internal vertex has at least one child; in particular, all unary portions of paths are retained. At each vertex, different children are drawn independently using the appropriate kernel. We write $E_{\mathcal{T}}$ for this sampling expectation conditional on the root. A split at depth d means that the last common vertex is at depth d . Replica positions at depth L are distinct vertices, but the values sampled there need not differ. Thus independent spin replicas may be the same spin configuration. Removing some leaves and the portions of their paths unused by the remaining leaves preserves the sampling law of the remaining tree.

When the terminal variables are spin configurations $\sigma^a \in \{-1, 1\}^N$, the multioverlap on a subset S of the replica leaves is

$$R_S = \frac{1}{N} \sum_{i=1}^N \prod_{a \in S} \sigma_i^a, \quad R_{\emptyset} = 1. \quad (3.1)$$

An unspecified expectation $\mathbb{E}$ below includes the root and the sampling of the indicated tree.

Here is how the path kernels arise from a partition function. Let P_d , $1 \leq d < L$, be the auxiliary prior kernels, put $X_{L-1} = \log Z$, and recursively set

$$X_{d-1} = m_d^{-1} \log P_d e^{m_d X_d} \quad (1 \leq d < L).$$

The corresponding *tilted kernels* are

$$K_d(dz \mid \mathcal{F}_{d-1}) = e^{m_d(X_d(z) - X_{d-1})} P_d(dz \mid \mathcal{F}_{d-1}), \quad 1 \leq d < L. \quad (3.2)$$

They are probability kernels by the recursion. The final kernel K_L is the Gibbs distribution of the spin configuration conditional on its auxiliary prefix. When $L = 1$, only this final kernel is present.

For a positive path function A , define

$$Y_L = A, \quad Y_{d-1} = (K_d Y_d^{m_d})^{1/m_d}, \quad d = L, \dots, 1. \quad (3.3)$$

We initially suppose that A is bounded above and away from zero, conditionally on the root. Every expression below is then well defined. Denote the kernels after insertion by

$$K_d^A(dz \mid \mathcal{F}_{d-1}) = \left(\frac{Y_d(z)}{Y_{d-1}} \right)^{m_d} K_d(dz \mid \mathcal{F}_{d-1}). \quad (3.4)$$

The recursion shows that these are probability kernels. Superscript A on a tree expectation means that these kernels are used.

Lemma 3.1 (Insertion calculus). *Suppose a leaf partition function is iterated by power means, with exponents $m_1, \dots, m_{L-1}$, and $K_1, \dots, K_{L-1}$ are its tilted kernels. Let K_L be its terminal Gibbs kernel. Multiplying each spin weight by A changes the iterated root logarithm by $\log Y_0$, where (3.3) uses the kernels before insertion. The new kernels are (3.4).*

For an internal vertex β of a prescribed tree, let d_β be its depth, let n_β be its number of children, and put

$$\gamma_\beta = m_{d_\beta} - n_\beta m_{d_\beta+1}.$$

The conditional joint density of its updated tree law is

$$W_{\mathcal{T}}(A) := \frac{dP_{\mathcal{T}}^A}{dP_{\mathcal{T}}} = \prod_{a \in \text{Leaves}(\mathcal{T})} A^a \prod_{\beta \in \text{Int}(\mathcal{T})} Y_{\beta}^{\gamma_{\beta}}. \quad (3.5)$$

Here Y_{β} is $Y_{d_{\beta}}$ evaluated on the prefix at β .

The recursion is monotone and homogeneous: if $a \leq A \leq b$, for positive root-measurable a, b , then $a \leq Y_d \leq b$ at every depth; and multiplying A by a positive root-measurable constant multiplies every Y_d by that constant.

Proof. At the terminal spin integration, the partition-function ratio is $K_L A = Y_{L-1}$, because $m_L = 1$. If X_d and X_d^A are the old and new backward logarithms, substitution in the preceding power mean gives

$$\exp(m_d(X_d^A - X_{d-1})) = K_d \exp(m_d(X_d^A - X_d)).$$

Backward induction proves $X_d^A - X_d = \log Y_d$ and then gives (3.4); the same formula at $d = L$ is the usual Gibbs density ratio.

Multiply the edge density ratios in (3.4) over the tree. An internal vertex at positive depth d receives exponent m_d from its incoming edge and exponent $-n_\beta m_{d+1}$ from its outgoing edges. At the root there is no incoming edge, which agrees with $m_0 = 0$. Each terminal vertex receives exponent $m_L = 1$. This proves (3.5). Monotonicity and homogeneity follow at once by backward induction in (3.3). $\square$

An insertion may use additional variables that were absent from the original model. Enlarge each path kernel by their independent prior kernels before applying the lemma. The original backward logarithms do not depend on these variables, so the original tilt leaves their conditional prior kernels unchanged. Additional root variables remain quenched.

We will use differentiation of (3.3). For $A_t = 1 + \sum_{j=1}^k t_j D_j$, where the real path functions D_j are bounded, differentiation on any real neighborhood where A_t is uniformly positive gives

$$\partial_{t_j} \log Y_d = E^{A_t} \left[\frac{D_j}{A_t} \middle| \mathcal{F}_d \right]. \quad (3.6)$$

The conditional expectation on the right follows one new path below the specified prefix. Indeed, differentiating one power mean gives $\partial_{t_j} \log Y_{d-1} = K_d^{A_t} \partial_{t_j} \log Y_d$, and the terminal derivative is D_j/A .

Analytic bounds. The expressions are analytic on the complex domain

$$\sum_j |t_j| \|D_j\|_\infty < 1,$$

with bounds uniform over the kernels. On a smaller closed domain with this sum at most $\rho < 1$, A has argument bounded by $\vartheta = \arcsin \rho < \pi/2$ and modulus between $1 - \rho$ and $1 + \rho$. If Y_d has argument in $[-\vartheta, \vartheta]$, then $K_d Y_d^{m_d}$ has argument in $[-m_d \vartheta, m_d \vartheta]$, using the principal powers. It is nonzero, and taking its $1/m_d$ power gives the same argument bound for Y_{d-1} . More quantitatively, a lower modulus bound $a_d > 0$ gives the lower bound $a_{d-1} = a_d \cos(m_d \vartheta)^{1/m_d} > 0$, whereas the upper modulus bound $1 + \rho$ is preserved. These bounds justify integration of analytic functions at each step and all subsequent differentiations. They also bound (3.5), its canceled versions below, and their derivatives on smaller domains, uniformly over the kernels for fixed depth, exponents, tree size, and bounds on the inserted functions.

3.2 The two expansion rules

Call a leaf *protected* if its factor A^a in the tree density is canceled by a denominator A^a . For a set C of protected leaves and a bounded parameter-independent function G , consider $E_{\mathcal{T}}^A[G \prod_{a \in C} (A^a)^{-1}]$. Differentiating (3.5) shows that

$$\partial_{t_j} \log W_{\mathcal{T}}(A) = \sum_{a \in \text{Leaves}(\mathcal{T})} \frac{D_j^a}{A^a} + \sum_{\beta \in \text{Int}(\mathcal{T})} \gamma_\beta \partial_{t_j} \log Y_\beta.$$

In the derivative of $E_{\mathcal{T}}^A[G \prod_{a \in C} (A^a)^{-1}]$, the leaf terms at $a \in C$ cancel exactly with differentiation of their denominators. Each remaining leaf term inserts D_j^a/A^a and makes that leaf protected. By (3.6), an internal-vertex term is the same expectation after adjoining an independent updated path below β , with an additional factor D_j/A at its terminal vertex. This is precisely a fresh path at β , with coefficient γ_β , and its new leaf is protected. Conditional independence of children and the projectivity of tree sampling justify this representation even when G uses the other children of β .

For a tree with n leaves, telescoping gives the useful identity

$$\sum_{\beta \in \text{Int}(\mathcal{T})} (n_\beta m_{d_\beta+1} - m_{d_\beta}) = n. \quad (3.7)$$

All summands are positive. Define a probability law Q_k on trees with leaves labeled $1, \dots, k$ as follows. Start with a single path. When the current tree has n leaves, choose an internal vertex β with probability

$$\frac{n_\beta m_{d_\beta+1} - m_{d_\beta}}{n}, \quad (3.8)$$

create a fresh child there, and continue with a completely new path to depth L . Give its terminal vertex label $n+1$. Identity (3.7) verifies the normalization. The random tree is chosen independently of all variables subsequently sampled on it. For example, under Q_2 the two paths split at depth d with probability $m_{d+1} - m_d$, for $0 \leq d < L$. This split is uniform on the depths when $m_d = d/L$.

Lemma 3.2 (Logarithmic expansion). *For a bounded real path function D and $A = 1 + tD$,*

$$\log Y_0 = \sum_{k \geq 1} \frac{(-1)^{k-1} t^k}{k} \mathbb{E}_{Q_k} E_{\mathcal{T}} \prod_{a=1}^k D^a, \quad |t| \|D\|_{\infty} < 1. \quad (3.9)$$

This identity is conditional on the root. If $|t| \|D\|_{\infty} \leq c < 1$, the absolute value of the remainder after order K is at most

$$\sum_{k > K} \frac{c^k}{k}. \quad (3.10)$$

Proof. The first derivative is, by (3.6), the expectation of D/A on one updated path. Protect that leaf. Every subsequent derivative can only add a fresh path, because all current leaves are protected. At order k , all $k - 1$ fresh coefficients are negative. After normalization by (3.7), their absolute products are $(k - 1)!$ times the probabilities in Q_k . Therefore

$$\left. \frac{d^k}{dt^k} \log Y_0 \right|_{t=0} = (-1)^{k-1} (k - 1)! \mathbb{E}_{Q_k} E_{\mathcal{T}} \prod_{a=1}^k D^a.$$

Analyticity proves (3.9); the coefficients have absolute value at most $\|D\|_{\infty}^k / k$, which proves the stated convergence domain and (3.10). $\square$

Here is the companion rule with old replicas. Fix a tree $\mathcal{T}$ with an anchor leaf a_* . Starting from all its old paths, assign colors $1, \dots, k$ in that order. For the next color, one may either

- (i) choose an unused old leaf other than a_* , with coefficient 1; or
- (ii) create a fresh path from any internal vertex β of the current tree, with coefficient $\gamma_{\beta} = m_{d_{\beta}} - n_{\beta} m_{d_{\beta}+1}$.

Each fresh path uses a new child at its divergence vertex and new vertices at every greater depth. Every chosen leaf receives its color and cannot be chosen again. The current tree includes all old paths, including unused ones, and every path already added. An extension history E records these choices; write $a_j(E)$ for the leaf carrying color j , $\mathcal{T}_E$ for its final tree, and c_E for the product of its coefficients.

Lemma 3.3 (Canceled-anchor expansion). *Let H be a bounded function of the variables on the old tree, fixed as the insertion parameters vary. For $A_t = 1 + \sum_{j=1}^k t_j D_j$, with bounded real path functions D_j , one has*

$$\partial_{t_1} \cdots \partial_{t_k} E_{\mathcal{T}} \frac{H}{A_t^{a_*}} \Big|_{t=0} = \sum_E c_E E_{\mathcal{T}_E} \left[H \prod_{j=1}^k D_j^{a_j(E)} \right]. \quad (3.11)$$

The formula holds in any fixed order of the colors. For identical directions $D_j = D$, the coefficient of t^k is the right-hand side divided by $k!$. The analytic bounds established above apply to its left-hand side, uniformly over the path kernels. The empty extension has weight 1.

Proof. Begin with $C = \{a_*\}$ and $G = H$. Apply the preceding rule successively, incorporating each new D_j in G . Fresh leaves are protected immediately, so the only unprotected leaves are unused old leaves. At zero all factors A and Y are 1, proving the claimed rule and its coefficients. $\square$

3.3 Sums of extension coefficients

We now control the signed normalization and the accumulated error in the prescribed-tree identities used for the lower bound.

In an extension history, the anchor and the already colored paths are called reference paths. Constraints on the new color will only specify its splitting depths with these paths. In particular, a constraint does not distinguish unused old paths with the same specified ancestry.

To see the cancellation that must be controlled, start with two old paths splitting at depth d , and let the first be the anchor. Require one added color to split from the anchor exactly at d . It may use the second old leaf, create a fresh path at the depth- d vertex, or follow the second old path and leave it at a depth $\ell > d$. The respective coefficients give the total

$$1 + (m_d - 2m_{d+1}) + \sum_{\ell=d+1}^{L-1} (m_\ell - m_{\ell+1}) = m_d - m_{d+1}.$$

On the uniform mesh $m_d = d/L$, with $\delta = L^{-1}$, this signed total is $-\delta$, but the sum of absolute coefficients is $2 + \delta$. If instead the required split is delayed to $d + 1 < L$, the only allowed choice is a fresh split at the unary depth- $(d + 1)$ vertex on the anchor path. Its coefficient is $-\delta$, so the absolute sum is now δ . The next two lemmas extend these two calculations: the first computes signed totals, and the second controls absolute sums when specified splits occur away from old branching depths.

Lemma 3.4 (Reference-path totals). *Suppose the next color must follow a specified reference prefix to depth d and then enter a child distinct from all $j \geq 1$ children there that contain reference paths. Apart from these ancestry constraints its continuation is unrestricted. The sum of the coefficients of all allowed choices is*

$$m_d - jm_{d+1}. \quad (3.12)$$

Consequently, if the anchor and the colors must have a prescribed tree S , the total coefficient over all compatible extensions of any old tree is

$$\kappa(S) = \prod_{v \in \text{Br}(S)} \prod_{j=1}^{a_v-1} (m_{d_v} - jm_{d_v+1}), \quad (3.13)$$

where a_v and d_v are the child count and depth of each branching vertex in the prescribed reference tree. More generally, the same one-step rule can be iterated starting from any already constructed set of references.

Proof. At the specified divergence vertex, let n be the child count in the entire current tree. The j excluded children contain all references passing through that vertex. Each of the other $n - j$ children roots a subtree containing no reference. Its terminal vertices are therefore unused old leaves. If such a subtree is rooted at depth $d + 1$, its permitted choices are an old leaf, with coefficient 1, or a fresh divergence at an internal vertex in the subtree. If it has q leaves, telescoping within it gives

$$q + \sum_{\beta \text{ internal in the subtree}} (m_{d_\beta} - n_\beta m_{d_\beta+1}) = m_{d+1}.$$

The formula includes the case of a terminal child, when both sides are 1. Adding these contributions to the fresh choice at the specified vertex gives $m_d - nm_{d+1} + (n - j)m_{d+1} = m_d - jm_{d+1}$.

For consistent prescribed splitting depths with all reference paths, the deepest required common prefix determines the divergence vertex and its excluded reference children. Thus the one-step sum depends only on the current reference tree, not on which old paths or fresh paths realized it. Sum backwards over successive choices. At a branching vertex of the final prescribed tree, the numbers of excluded reference children when new child groups are first entered are exactly $1, \dots, a_v - 1$. Their factors give (3.13), independently of the color order. $\square$

To control errors after division by a small reference coefficient, we also need an absolute bound. Moving a target branching depth away from all old branching depths forces a fresh unary split. The next estimate charges one small factor to each such vertex.

Lemma 3.5 (Charging fresh splits). *Assume $m_d = d/L$ and put $\delta = L^{-1}$. Start with an old tree having q leaves and assign k colors by the canceled-anchor extension rule. Prescribe the reference shape of the anchor and the colors, possibly with additional restrictions on allowed extensions. Suppose b specified branching vertices of that shape have depths which are not branching depths anywhere in the old tree. Then*

$$\sum_{E \text{ allowed}} |c_E| \leq C_{q,k} \delta^b. \quad (3.14)$$

The constant is independent of L , the prescribed depths, and their coincidences at different vertices.

In particular, for these b vertices let

$$J = \prod_v \prod_{j=1}^{a_v-1} (m_{d_v} - j m_{d_v+1}).$$

If $m_{d_v} \geq \eta > 0$ at all of them, then $J \neq 0$ and

$$\frac{1}{|J|} \sum_{E \text{ allowed}} |c_E| \leq C_{q,k,\eta}. \quad (3.15)$$

Proof. If the current tree has s leaves, (3.7) says that the total absolute coefficient of all fresh choices is s . There are at most s eligible old choices. Hence the total absolute coefficient at an unrestricted step is at most $2s \leq 2(q+k)$. At any fixed depth there are at most s vertices. Restricting a step to fresh divergences at unary vertices of a prescribed depth therefore gives total absolute coefficient at most $s\delta$, because each such divergence has coefficient $m_d - m_{d+1} = -\delta$.

Consider an allowed history and one of its specified target branching vertices. That vertex cannot already be branching in the old tree. At the first step at which it became branching in the current tree, a fresh path must therefore have diverged there while it was unary. The vertex may itself have been created earlier as part of a new unary path; the same conclusion then applies. Distinct target vertices become branching at distinct steps, since a fresh path creates only one branching vertex. Assign each of the b vertices to its creation step. There are at most $k!/(k-b)!$ possible assignments (and necessarily $b \leq k$).

For each fixed assignment, enlarge the family of histories by discarding all constraints except that a designated step creates a unary split at its assigned depth. Bound its step sum by $(q+k)\delta$ and every other step sum by $2(q+k)$. Iterated summation over choices gives at most

$$\frac{k!}{(k-b)!} (q+k)^b (2(q+k))^{k-b} \delta^b.$$

This proves (3.14), including repeated assigned depths, because the designated steps remain distinct.

For the final assertion, the $j = 1$ factor at each designated vertex has absolute value δ , whereas for $j \geq 2$

$$|m_d - jm_{d+1}| = (j-1)m_d + j\delta \geq \eta.$$

There are only a size-dependent number of the latter factors. Thus $|J| \geq \delta^b \eta^{\sum_v (a_v - 2)}$, and the result follows. $\square$

3.4 The distribution of branching depths

Still take $m_d = d/L$. Suppress unary stretches only for the purpose of describing a shape by a finite topology and its branching-depth coordinates. For $0 < \eta < 1/2$, call a shape *regular* if every branching depth divided by L lies in $(\eta, 1 - \eta)$ and any two distinct branching vertices have normalized depths differing by more than η . A singleton is regular. This is the regularity used when sampling a single tree; later we also use two copies whose corresponding depth assignments are deliberately aligned.

Lemma 3.6 (Regular shapes). *For every fixed k there is $C_k < \infty$, independent of L and η , such that*

$$Q_k\{\mathcal{T} \text{ is not regular}\} \leq C_k(\eta + L^{-1}). \quad (3.16)$$

For any particular labeled topology with a branching vertices and any consistent assignment of their depths,

$$Q_k\{\text{that topology and depth assignment}\} \leq C_k L^{-a}. \quad (3.17)$$

Deleting leaves from a regular tree produces a regular tree, after discarding branching vertices that become unary.

Proof. Suppose the current tree has n leaves. A new branching vertex is created exactly when the next path diverges at a unary vertex. Each such choice has probability δ/n by (3.8). At any depth there are at most n vertices; hence for a set B of depths the probability that the next path creates a branching vertex at a depth in B is at most $|B|\delta$. At each of the first $k-1$ steps, the boundary depths and the depths within normalized distance η of any previous branching vertex form a set of cardinality at most $C_k(\eta L + 1)$. If none of these forbidden creation events occurs, the final tree is regular. A union bound proves (3.16). Divergence at an already branching vertex does not create a new coordinate and requires no extra exclusion.

For (3.17), restrict a specified final labeled tree to leaves $1, \dots, n$, retaining every unary stretch. These restrictions determine its successive histories under (3.8). Each of its a branching vertices is first created by a unary divergence, whose probability is at most δ ; every other transition probability is at most 1. The probability of the specified history is therefore at most δ^a . If a topology description identifies equivalent child orderings, their number is bounded in terms of k alone, which gives the stated version with C_k . Finally, a retained branching vertex after deletion was already a branching vertex of the original tree, at the same depth. Its depth coordinates are a subset of the old ones, so regularity is preserved. $\square$

Remark 3.7. For fixed k there are finitely many labeled topologies. Therefore (3.17) converts an unnormalized depth average $L^{-a} \sum$ for a topology with a branching coordinates into a bound for its contribution under Q_k . The same is true for errors on subshapes: summing over coordinates discarded by the subshape costs at most the corresponding power of L . Together with (3.16), this allows averaged estimates on regular depth assignments to be used in the logarithmic expansion.

4 The interpolation upper bound

The interpolation method of Guerra [11], adapted to diluted systems by Franz and Leone [9] and to arbitrary finite levels by Panchenko and Talagrand [16], replaces each real interaction by p independent cavity interactions. The difference is controlled by the convexity of $x \mapsto x^p$. We give the argument for the full trial class, including unbounded messages and integrable physical disorder.

Proposition 4.1. *Under the hypotheses of Theorem 2.1, $F_N \leq \mathcal{B}_r(\zeta, \mathbf{m})$ for every N and every admissible finite-level trial law. Consequently $F_N \leq \inf_{r \geq 0} \Phi_r$.*

Proof. First dispose of the degenerate factorization. Put $M = \max_{s=\pm 1} |f(s)|$. If $b = 0$ almost surely or $M = 0$ almost surely, the interaction is a spin-independent random constant c , and

$$F_N = \mathcal{B}_r = \log 2 + \mathbb{E} \log \cosh h + \alpha \mathbb{E} c$$

for every trial law. Otherwise choose $u, v > 0$ with $\mathbb{P}(M \geq u) > 0$ and $\mathbb{P}(|b| \geq v) > 0$. Since $|b| \prod_{\ell=1}^p M_\ell < 1$ almost surely, independence gives

$$\|b\|_\infty \leq u^{-p}, \quad \|M\|_\infty \leq (vu^{p-1})^{-1}.$$

Indeed, violating either bound on a set of positive probability and restricting the other independent factors to these lower bounds would violate the strict product condition. All moments used below are therefore finite.

Interpolation. Fix a trial law and set $L = r + 1$. Interpolate between real interactions of mean $t\alpha N$ and, at each site i , independent cavity interactions $U_\theta(x_1, \dots, x_{p-1}; \sigma_i)$ of mean $(1-t)\alpha p$. Different cavity terms have disjoint message labels. Counts, interactions, fields, and root message measures are quenched. Apply the backward power-mean recursion to the spin partition function and denote its expected root logarithm divided by N by $\varphi(t)$. For $r = 0$ all messages are quenched and there is no auxiliary recursion. Independence between different sites throughout the recursion gives

$$\varphi(1) = F_N, \quad \varphi(0) = \log 2 + \mathbb{E} \log T_0 V_s.$$

Poisson differentiation gives $\varphi'(t)/\alpha$ as the expected insertion increment of one real interaction minus p times the increment of one cavity interaction at a uniform site. Each increment has absolute value at most $B(\theta)$, since the recursion preserves multiplicative bounds. Conditioning on the counts justifies the derivative on $(0, 1)$, and the same bound gives continuity at the endpoints.

We claim that

$$\varphi'(t) + \alpha(p-1)\mathbb{E} \log T_0 V_e \leq 0. \tag{4.1}$$

All three insertions have the form $a(1+D)$ under (2.3). For a real interaction, $D = b \prod_\ell f_\ell(\sigma_{i_\ell})$; for a cavity interaction, replace $p-1$ of the factors by $\bar{f}(x) = \sum_s f(s)q_x(s)$; for an edge insertion, replace all p . In the last case the independent message process is appended to the unperturbed tilted tree, so its increment is $\log T_0 V_e$.

Normalization. We remove a before taking expectations, so no integrability of $\log a$ is needed. Put $D_* = b \prod_\ell f_\ell(1)$ and $c = \theta(1, \dots, 1)$. Subtract c from each insertion increment. These integrable subtractions cancel in the combination with coefficients $1, -p, p-1$. Keep the interpolated system fixed. For $0 < \lambda < 1$, replace each normalized increment by

$$\log Y_0(1 + \lambda D) - \log(1 + \lambda D_*), \tag{4.2}$$

where Y_0 is the insertion recursion of [Lemma 3.2](#). For every spin input, the ratio

$$\frac{1 + \lambda D}{1 + \lambda D_*} = (1 - w) + w \frac{1 + D}{1 + D_*}, \quad w = \frac{\lambda(1 + D_*)}{1 + \lambda D_*} \in [0, 1],$$

lies between $e^{-2B(\theta)}$ and $e^{2B(\theta)}$. The same is true after averaging any subset of the spin inputs against the q_x 's. It follows that (4.2) is bounded in absolute value by $2B(\theta)$. For each fixed interaction, convergence as $\lambda \uparrow 1$ is uniform in the spin inputs and messages and is preserved by the finite recursion. Dominated convergence therefore recovers the desired normalized increments.

Convexity. Apply [Lemma 3.2](#) to the regularized increments. On a tree $\mathcal{T}$ with n replica leaves define

$$P_{\mathcal{T}} = \frac{1}{N} \sum_{i=1}^N \mathbb{E}_f \prod_{a=1}^n f(\sigma_i^a), \quad B_{\mathcal{T}} = \mathbb{E}_{f,x} \prod_{a=1}^n \bar{f}(x^a).$$

Here $\mathbb{E}_f$ averages one fresh function f , shared by all n replica factors in the product. The expectation $\mathbb{E}_{f,x}$ also averages an entire fresh message-label tree, including its root. This function and message tree are independent of one another and of the old tilted system. Thus $B_{\mathcal{T}}$ is deterministic for a fixed tree, while $P_{\mathcal{T}}$ still depends on the old spin replicas. At $r = 0$, the message itself is a root variable shared by all replica leaves. Averaging the independent uniform indices and the independent copies of f and of the new message label gives, respectively, $P_{\mathcal{T}}^p$, $P_{\mathcal{T}} B_{\mathcal{T}}^{p-1}$, and $B_{\mathcal{T}}^p$. The inserted b is independent of these variables. Thus the order- n coefficient of the combined regularized increment is

$$-\frac{\lambda^n}{n} \mathbb{E}[(-b)^n] \mathbb{E}_{Q_n} \mathbb{E} \left[P_{\mathcal{T}}^p - p P_{\mathcal{T}} B_{\mathcal{T}}^{p-1} + (p-1) B_{\mathcal{T}}^p \right]. \quad (4.3)$$

The regularized baseline terms cancel. The tree law Q_n is that of [Lemma 3.2](#); it is common to the three insertions. The series is absolutely convergent for $\lambda < 1$, with each insertion controlled by $\sum_{n \geq 1} \lambda^n / n$. Since p is even, convexity of $x \mapsto x^p$ makes the bracket nonnegative, and (2.4) makes (4.3) nonpositive. Taking $\lambda \uparrow 1$ proves (4.1). Integration over $t \in [0, 1]$ gives the proposition. $\square$

5 A perturbed cavity reservoir

We compare systems of sizes N and $N + 1$ through a common N -spin system, the *reservoir*. This cavity comparison follows the Aizenman–Sims–Starr variational scheme [2] and its spin-distribution form in Panchenko [14, Section 2.2]. We add a sublinear number of bounded perturbations to obtain identities for its spin trees. The comparison below shows that these perturbations change the pressure by $o(1)$ and preserve the site-minus-bond cavity increment.

Until [Section 9](#), assume that $B(\theta) \leq B$ and $|h| \leq H$ almost surely and that the law of θ is invariant under permutations of its p coordinates. Neither the factorization nor the positivity assumption is used in the lower bound.

Fix $L \geq 2$ and use $r = L - 1$ auxiliary levels with

$$m_d = d/L \quad (0 \leq d \leq L), \quad \delta = 1/L. \quad (5.1)$$

Let $(c_\nu)_{\nu \geq 1}$ be positive with $\sum_\nu c_\nu = 1$, and let $s_N = N^{3/4}$. To the spin weight add independent Poisson families of single-site factors ψ_ν , with counts of means $c_\nu s_N$ and independent uniform sites in $[N]$. Each factor uses its own auxiliary variables, drawn along the hierarchy with product priors, and satisfies

$$1/2 \leq \psi_\nu \leq 3/2. \quad (5.2)$$

The total number of factors is $\text{Poisson}(s_N)$ and is finite almost surely. Counts, types, and sites are root variables. The parameters of the factors are denoted by $\mathbf{u}$ and are fixed, not quenched random variables, unless a parameter average is explicitly taken. The factors needed later are specified in [Section 6](#).

Apply the power-mean recursion to the resulting leaf partition function. Write $P_N^{L,\mathbf{u}}$ for its expected root logarithm. Comparing pointwise with the unperturbed spin weight and then applying the recursion gives

$$|P_N^{L,\mathbf{u}} - NF_N| \leq (\log 2)s_N, \quad (5.3)$$

uniformly in the hierarchy, its parameters, and N .

Lemma 5.1 (Cavity increments). *The increments $\Delta_N^{L,\mathbf{u}} = P_{N+1}^{L,\mathbf{u}} - P_N^{L,\mathbf{u}}$ are uniformly bounded in $N, L, \mathbf{u}$. There is a reservoir on N spins with physical interaction count of mean*

$$\rho_N = \alpha(N+1) \left(\frac{N}{N+1} \right)^p, \quad (5.4)$$

the usual fields, and the perturbation counts of means $c_\nu s_N$, such that

$$\Delta_N^{L,\mathbf{u}} = \log 2 + \mathbb{E} \log Y_{0,s} - \mathbb{E} \log Y_{0,b} + o_N(1). \quad (5.5)$$

The error is uniform in $L, \mathbf{u}$. The insertion recursion uses the tilted reservoir kernels and independent new root disorder, with terminal insertions

$$A_s(\sigma) = \frac{1}{2} \sum_{\varepsilon=\pm 1} \exp \left(h\varepsilon + \sum_{j=1}^{K_s} \theta_j(\sigma_{i_{1,j}}, \dots, \sigma_{i_{p-1,j}}, \varepsilon) \right), \quad (5.6)$$

$$A_b(\sigma) = \exp \left(\sum_{j=1}^{K_b} \theta_j(\sigma_{i_{1,j}}, \dots, \sigma_{i_{p,j}}) \right), \quad (5.7)$$

where $K_s \sim \text{Poisson}(\alpha p)$, $K_b \sim \text{Poisson}(\alpha(p-1))$, and all displayed indices are independent uniform indices in $[N]$.

Proof. The physical interactions. Split the Poisson interaction process of the $(N+1)$ -spin system according to the number of indices equal to $N+1$. Terms with no new index have mean ρ_N . Terms with exactly one new index have mean $\alpha p(N/(N+1))^{p-1} = \alpha p + O(N^{-1})$; by permutation invariance we may place the new input last. Their other indices are independent uniform indices in $[N]$, and their interaction functions retain the prescribed law. Terms with at least two new indices have total mean $O(N^{-1})$. Meanwhile, the N -spin system can be obtained from the same reservoir by adding a Poisson number of old interactions of mean

$$\alpha N - \rho_N = \alpha(p-1) + O(N^{-1}).$$

These quantities are nonnegative for all sufficiently large N .

The perturbations. The old-site perturbation count in the $(N+1)$ -spin system has total mean $s_{N+1}N/(N+1)$, and the new-site count has mean $s_{N+1}/(N+1)$. Couple these processes to the reservoir perturbations using common Poisson points. The expected number of unmatched factors is at most

$$|s_{N+1} - s_N| + 2s_{N+1}/(N+1) = O(N^{-1/4}).$$

Deleting or adding a perturbation changes the root logarithm by at most $\log 2$; an interaction changes it by at most B . An unused auxiliary variable can simply be integrated out with its prior, so these comparisons remain valid when the number of auxiliary variables changes.

The increment. We may consequently discard the repeated-new-index interactions, match the perturbations, and replace the two insertion count means by their limits at a total expected cost $O(N^{-1/4})$. Summing over the new spin gives $2A_s$, while the old-system correction gives A_b . The factor 2 supplies the term $\log 2$ in the increment. The insertion identity for the power means then proves (5.5). Finally,

$$|\log Y_{0,s}| \leq H + BK_s, \quad |\log Y_{0,b}| \leq BK_b.$$

Together with the coupling estimate this bounds the increments for large N ; increasing the constant covers the remaining sizes. All estimates are uniform in L and $\mathbf{u}$. $\square$

In the rest of the lower-bound proof, all spin trees are sampled from this reservoir using its tilted auxiliary kernels and its terminal Gibbs kernel. The reservoir retains an independent Poisson interaction process of mean $O(N)$ and N independent bounded fields. Thus the concentration argument applies to its slightly modified interaction mean as well.

6 Perturbations and identities on prescribed trees

Throughout this section the interaction and field are bounded, and the depth $L \geq 2$ is fixed. The arguments apply both to the N -spin system and to the reservoir in Lemma 5.1. Its physical disorder consists of independent bounded interactions with a Poisson count of mean $O(N)$, together with N independent bounded fields. All expectations in a statement at a specified parameter vector are taken with that vector held fixed.

We choose perturbations whose derivatives concentrate, while selecting cavity increments at most an arbitrarily small error above the unperturbed liminf. A Poisson insertion then turns concentration into Taylor-coefficient identities. Independent signs along the inserted paths select the prescribed splitting depths. The use of concentration to obtain overlap identities has antecedents in the self-averaging argument of Ghirlanda–Guerra [10]. Related antecedents are the stochastic-stability relations of Aizenman–Contucci [1] and their extensions to diluted multioverlaps by Barra and De Sanctis [3]. Panchenko’s spin-distribution identities [14] provide the cavity context. The bounded perturbation family and the identities on colored finite trees needed here are constructed and proved below.

6.1 The perturbation family

A *mark specification* consists of a finite number $n \geq 0$ of labeled Rademacher coordinates, a depth in $\{1, \dots, L-1\}$ assigned to each coordinate, two functions $g, g_0 : \{-1, 1\}^n \rightarrow \mathbb{Q} \cap [-1, 1]$, and a choice $S_0 = 1$ or $S_0 = \sigma_i$. Each coordinate is sampled at its assigned depth along an auxiliary path. Distinct coordinates are independent; the value of a coordinate is shared at a shared vertex and is independent at different vertices. The case $n = 0$ includes constant functions. We use all such specifications, a countable collection because each sign domain is finite. Products of signs and their rational averages belong to this collection; these will select branching patterns.

The two parameters have different roles: differentiation in u produces a concentrated score, while the values of t probe an insertion’s Taylor expansion. Taking u smaller than every fixed power of t will force each limiting Taylor coefficient to vanish. For each specification include types indexed by all integers $j \geq 4$. The type ν corresponding to this specification and j has

$$t_\nu = j^{-1}, \quad I_\nu = [j^{-j}, 2j^{-j}], \quad u_\nu \in I_\nu. \quad (6.1)$$

Enumerate these types by $\nu \in \mathbb{N}$ and choose constants $c_\nu > 0$ with $\sum_\nu c_\nu = 1$. Independently for each type, put $n_\nu \sim \text{Poisson}(c_\nu s_N)$, where $s_N = N^{3/4}$. Each term has an independent uniform site $i \in [N]$ and an independent copy of its mark process. At a leaf it multiplies the spin weight by

$$\psi_\nu = 1 + t_\nu g \sigma_i + u_\nu g_0 S_0. \quad (6.2)$$

In particular $1/2 < \psi_\nu < 3/2$. Counts, types, and sites belong to the root; the mark variables are sampled with their product priors at their assigned auxiliary depths. Since the total count is $\text{Poisson}(s_N)$, this construction uses only finitely many terms almost surely. The factor bound permits termwise comparison through the power means, as in (5.3).

Write π for the product of the uniform probability measures on the intervals I_ν , and $\mathbf{u} = (u_\nu)_\nu$. We use π to select deterministic parameter vectors, not as part of their quenched law. Each interval has fixed positive length for a fixed type; concentration estimates may therefore depend on its length.

6.2 Concentration of each perturbation score

For one type ν , define on a tilted path, including its terminal Gibbs configuration,

$$D_\nu = \sum_{q=1}^{n_\nu} \frac{g_{0,q} S_{0,q}}{1 + t_\nu g_q \sigma_{i_q} + u_\nu g_{0,q} S_{0,q}}. \quad (6.3)$$

Let $\langle \cdot \rangle$ denote expectation along one tilted path conditional on the root, and let $\mathbb{E}_{\mathbf{u}}$ average the root at fixed $\mathbf{u}$. Define

$$e_{\nu,N}(\mathbf{u}) = \frac{1}{c_\nu s_N} \mathbb{E}_{\mathbf{u}} \langle |D_\nu - \mathbb{E}_{\mathbf{u}} \langle D_\nu \rangle| \rangle. \quad (6.4)$$

Lemma 6.1 (Averaged score concentration). *For each fixed L and each fixed type ν ,*

$$\int e_{\nu,N}(\mathbf{u}) \pi(d\mathbf{u}) \longrightarrow 0. \quad (6.5)$$

The assertion holds for the reservoir as well as for the original system. Constants need not be uniform in L or ν .

Proof. Hold the other parameters fixed, write $u = u_\nu$, $n = n_\nu$, and put $\lambda_\nu = c_\nu s_N$. Choose an open interval J_ν containing I_ν such that $|t_\nu| + |u| < 1/2$ throughout its closure. On this interval, pathwise,

$$|D_\nu| \leq 2n, \quad -4n \leq \partial_u D_\nu = - \sum_{q=1}^n \frac{(g_{0,q} S_{0,q})^2}{\psi_{\nu,q}^2} \leq 0. \quad (6.6)$$

Fluctuations along a tilted path. Let $X_0(u)$ be the random root log partition function before taking the root expectation. To include the last step in the differentiation, regard the logarithm of the spin weight as X_L and the spin partition function as the step with exponent $m_L = 1$. Differentiation of one backward step gives

$$\partial_u X_{d-1} = K_d(\partial_u X_d), \quad \partial_u^2 X_{d-1} = K_d(\partial_u^2 X_d) + m_d \text{Var}_{K_d}(\partial_u X_d).$$

Thus, for $M_d = \langle D_\nu \mid \mathcal{F}_d \rangle$,

$$X'_0(u) = \langle D_\nu \rangle, \quad (6.7)$$

$$\begin{aligned} X''_0(u) &= \langle \partial_u D_\nu \rangle + \sum_{d=1}^L m_d \langle (M_d - M_{d-1})^2 \rangle \\ &\geq m_1 \text{Var}(D_\nu \mid \mathcal{F}_0) - 4n. \end{aligned} \quad (6.8)$$

Here the conditional expectations and variances use the tilted path law at the current u . At fixed root all derivatives are justified by the bounded score and its derivatives; the finite recursion then justifies the displayed differentiation formulas. Their root expectations are justified by the finite moments of the Poisson counts.

Integrating (6.8) over $I_\nu = [a, b]$ and using $|X'_0| \leq 2n$ yields

$$\int_a^b \mathbb{E}_{\mathbf{u}} \text{Var}(D_\nu \mid \mathcal{F}_0) du \leq \frac{4\lambda_\nu(1+b-a)}{m_1}. \quad (6.9)$$

Consequently the interval-averaged thermal mean absolute deviation, divided by λ_ν , is at most

$$\left(\frac{4(1+|I_\nu|)}{m_1|I_\nu|\lambda_\nu} \right)^{1/2}. \quad (6.10)$$

Fluctuations of the root mean. It remains to concentrate X'_0 over the root. By (6.8), the random function

$$G(u) = X_0(u) + 2nu^2$$

is convex on J_ν . Uniformly for $u \in J_\nu$ and all the other parameters,

$$\text{Var}(G(u)) \leq C_\nu N, \quad |\mathbb{E}_{\mathbf{u}} G'(u)| \leq C_\nu \lambda_\nu. \quad (6.11)$$

To prove the first estimate, condition on the physical and total perturbation counts. Conditional on the latter count, its types and sites are independent labels; changing a label replaces a single bounded factor. Apply the Efron–Stein variance inequality [8] $\text{Var} Z \leq \frac{1}{2} \sum_j \mathbb{E}(Z - Z^{(j)})^2$, where $Z^{(j)}$ is obtained by independently resampling the j th input. For completeness, with independent inputs $X_1, \dots, X_k$, let E_j integrate only X_j and let $\mathcal{G}_j = \sigma(X_1, \dots, X_j)$. The Doob differences are $D_j = \mathbb{E}[Z - E_j Z \mid \mathcal{G}_j]$. Orthogonality and Jensen give

$$\text{Var} Z = \sum_j \mathbb{E} D_j^2 \leq \sum_j \mathbb{E}(Z - E_j Z)^2 = \frac{1}{2} \sum_j \mathbb{E}(Z - Z^{(j)})^2.$$

This independent-input decomposition requires neither identical laws nor a symmetric statistic; see Efron–Stein [8, Section 2]. Resampling one interaction, field, or perturbation term changes X_0 by a bounded amount: this is immediate for the spin weights and is preserved by each power mean. When a term is removed, its unused independent mark variables can be integrated out, so the same assertion holds for addition or deletion of a term. Conditional variances are therefore bounded by a constant times the number of root variables. Conditional means are Lipschitz functions of the counts, with bounded Lipschitz constants. If Q, Q' are independent copies of a Poisson variable and b is C -Lipschitz, then

$$\text{Var}(b(Q)) = \frac{1}{2} \mathbb{E}(b(Q) - b(Q'))^2 \leq C^2 \text{Var}(Q).$$

Applying this to the independent counts completes the $O(N)$ estimate. The correction $2nu^2$ obeys the same estimates. The derivative bound in (6.11) follows directly from (6.6).

Passing from a convex function to its derivative. Put $\bar{G} = \mathbb{E}_{\mathbf{u}} G$ and $Z = G - \bar{G}$. If $u \pm \rho$ lie in J_ν , forward and backward difference quotients imply

$$|G'(u) - \bar{G}'(u)| \leq \frac{|Z(u - \rho)| + 2|Z(u)| + |Z(u + \rho)|}{\rho} + \bar{G}'(u + \rho) - \bar{G}'(u - \rho).$$

The monotonicity of $\bar{G}'$ gives

$$\int_a^b [\bar{G}'(u + \rho) - \bar{G}'(u - \rho)] du \leq 2\rho[\bar{G}'(b + \rho) - \bar{G}'(a - \rho)].$$

Together with (6.11), this proves

$$\frac{1}{|I_\nu|} \int_{I_\nu} \mathbb{E}_{\mathbf{u}} |G'(u) - \mathbb{E}_{\mathbf{u}} G'(u)| du \leq C_\nu \left(\frac{\sqrt{N}}{\rho} + \frac{\rho \lambda_\nu}{|I_\nu|} \right). \quad (6.12)$$

For all sufficiently large N , take $\rho = N^{-1/8}$, which lies within the fixed neighborhood margin. After division by $\lambda_\nu = c_\nu N^{3/4}$, both terms tend to zero. Removing the correction to G' costs at most $C_\nu \mathbb{E}|n - \lambda_\nu|/\lambda_\nu \leq C_\nu \lambda_\nu^{-1/2}$. Thus the root contribution to (6.4) tends to zero. Combining it with (6.10) proves the result after integration over u_ν . All estimates are uniform in the other parameters, so their integration against the remaining product measure proves (6.5). $\square$

6.3 One sequence for low increments and all identities

The parameter choices must satisfy concentration and a low cavity increment simultaneously. Let $\underline{F} = \liminf_N F_N$ over all system sizes for the bounded model. Recall that $P_N^{L, \mathbf{u}}$ denotes the root-averaged perturbed log partition function, and put $\Delta_N(\mathbf{u}) = P_{N+1}^{L, \mathbf{u}} - P_N^{L, \mathbf{u}}$.

Lemma 6.2 (Deterministic parameter selection). *For every fixed $L \geq 2$ and $\varepsilon > 0$, there are sizes $N_j \uparrow \infty$ and deterministic vectors $\mathbf{u}_j$ such that*

- (i) $\Delta_{N_j}(\mathbf{u}_j) \leq \underline{F} + 2\varepsilon$;
- (ii) for every fixed type ν , $e_{\nu, N_j}(\mathbf{u}_j) \rightarrow 0$ in the reservoir;
- (iii) in the reservoir, the expectations of every finite product of multioverlaps on every fixed finite prescribed tree of depth L converge along this sequence.

Proof. Find low averaged increments. The uniform perturbation bound (5.3) gives

$$\liminf_N \frac{1}{N} \int P_N^{L, \mathbf{u}} \pi(d\mathbf{u}) = \underline{F}.$$

Hence infinitely many N satisfy $\int \Delta_N d\pi \leq \underline{F} + \varepsilon$: otherwise telescoping the averaged increments would contradict this liminf. Lemma 5.1 supplies a constant C such that $|\Delta_N(\mathbf{u})| \leq C$ uniformly.

Choose parameters with small score errors. For a finite collection of types, Lemma 6.1 and Markov's inequality show that the set where all its errors are small has parameter measure tending to one. More explicitly, choose N_j successively among the preceding infinite set so large that

$$\sum_{\nu \leq j} \int e_{\nu, N_j}(\mathbf{u}) \pi(d\mathbf{u}) \leq j^{-3}.$$

Then the set G_j where $e_{\nu, N_j} \leq j^{-1}$ for all $\nu \leq j$ has complement of measure $q_j \leq j^{-2}$. Its conditional mean increment is at most

$$\frac{\underline{F} + \varepsilon + Cq_j}{1 - q_j} \leq \underline{F} + 2\varepsilon$$

for all sufficiently large j . A vector in G_j with increment at most $\underline{F} + 2\varepsilon$ can therefore be chosen. Relabeling the tail proves (i) and (ii).

Make all finite-tree moments converge. At fixed L the finite labeled tree shapes and finite products of their multioverlaps form a countable collection. Each such product is bounded by one in absolute value. A diagonal subsequence gives (iii), without affecting (i) or (ii). $\square$

From now on L is fixed and $N \rightarrow \infty$ means convergence along a sequence from Lemma 6.2. Expectations below are evaluated at the selected deterministic vector, whose dependence on N is suppressed. No uniformity in L , in the number of paths, or in a perturbation type is asserted for these limits.

6.4 Poisson insertion and analytic extraction

Fix an old labeled tree $\mathcal{T}$, a distinguished leaf a_* , and a bounded polynomial f in multioverlaps of its old replicas. The polynomial and its coefficients are fixed as N and the insertion parameters vary. The root-averaged expectation on this tree is denoted simply by $\mathbb{E}f$. The anchor marginal is always one tilted path. Therefore (6.4) implies, for each fixed type ν ,

$$\left| \frac{\mathbb{E}[fD_\nu^{a_*}] - \mathbb{E}f \mathbb{E}\langle D_\nu \rangle}{c_\nu s_N} \right| \leq \|f\|_\infty e_{\nu, N} \rightarrow 0. \quad (6.13)$$

For a fixed mark specification, introduce an independent new uniform site and an independent copy of its mark process, and set $A = 1 + t g \sigma_i + u g_0 S_0$. Let Y be its insertion recursion from Lemma 3.1, and let $W_{\mathcal{T}}(A)$ be the updated-tree density (3.5). The expectations in the following display use the unmodified reservoir, extended by the independent insertion variables with their prior kernels:

$$J_{N, \mathcal{T}}(t, u) = \mathbb{E} \left[f \frac{g_0^{a_*} S_0^{a_*}}{A^{a_*}} W_{\mathcal{T}}(A) \right], \quad (6.14)$$

$$H_N(t, u) = J_{N, \mathcal{T}}(t, u) - \mathbb{E}f J_{N, \{a_*\}}(t, u). \quad (6.15)$$

In the single-anchor expression $J_{N, \{a_*\}}$, the test is 1.

Lemma 6.3 (Palm identity). *If ν is a type with the specified marks, then*

$$|H_N(t_\nu, u_\nu)| \leq \|f\|_\infty e_{\nu, N}. \quad (6.16)$$

For a fixed mark specification the function $H_N(t, u)$ is the same for all its types j : its base reservoir contains the full original Poisson families, including all types.

Proof. Apply the Poisson add-one identity to the sum over type- ν terms in D_ν . After division by its mean $c_\nu s_N$, the remaining count is again Poisson with mean $c_\nu s_N$, and all the other counts are unchanged. Thus the remaining model has exactly the original reservoir law. The distinguished

term becomes an independent new insertion with parameters t_ν, u_ν . Its score is $g_0^{a*} S_0^{a*} / A^{a*}$. The updated tree density is (3.5), so the anchor factor A^{a*} cancels. This proves

$$\frac{\mathbb{E}[f D_\nu^{a*}]}{c_\nu s_N} = J_{N, \mathcal{T}}(t_\nu, u_\nu), \quad \frac{\mathbb{E}\langle D_\nu \rangle}{c_\nu s_N} = J_{N, \{a*\}}(t_\nu, u_\nu).$$

Equation (6.16) now follows from (6.13). In particular no type has been deleted from the law of the base reservoir. This proves the last assertion. $\square$

Lemma 6.4 (Vanishing insertion coefficients). *For every mark specification included in the perturbation family, old tree, anchor, and bounded overlap-polynomial test f , every Taylor coefficient at $t = 0$ of $H_N(t, 0)$ tends to zero along the selected sequence.*

Proof. Lemma 3.3 gives a complex neighborhood $|t|, |u| < \rho$ on which the functions in (6.15) are analytic and uniformly bounded in N and in the selected parameter vector. Shrinking ρ gives a uniform bound on their u -derivatives as well. These constants may depend on the fixed tree, L , and $\|f\|_\infty$, but not on the type index j in the mark specification.

Each Taylor coefficient of $H_N(t, 0)$ is a finite linear combination of multioverlap moments and products of two such moments on prescribed trees. Their coefficients are determined by the fixed tree, exponents, mark law, and polynomial test, and are independent of N . Indeed the canceled anchor expansion supplies finitely many extensions at any fixed order; averaging the independent insertion site turns its spin product into a multioverlap. These coefficients converge by Lemma 6.2(iii). The uniform analytic bound then implies, by the geometric bound on the Taylor tails, that $H_N(t, 0)$ converges uniformly on each smaller disk to an analytic function $H(t)$.

For every sufficiently large fixed j , the corresponding type and Lemma 6.3 give

$$\limsup_{N \rightarrow \infty} |H_N(j^{-1}, 0)| \leq C \limsup_{N \rightarrow \infty} |u_{\nu, N}| \leq 2Cj^{-j}.$$

Here the score error tends to zero with j fixed, and the same function H_N is used for every j , by Lemma 6.3. Hence $|H(j^{-1})| \leq 2Cj^{-j}$. If the first nonzero Taylor coefficient of H had degree q , then $|H(j^{-1})| \geq cj^{-q}$ for all sufficiently large j , a contradiction. Thus H vanishes identically near zero, proving the claim coefficient by coefficient. $\square$

6.5 Prescribing the colored tree

The anchor is called a *spin anchor* when $S_0 = \sigma_i$, and a *marker anchor* when $S_0 = 1$. Added colors are ordered $1, \dots, k$; the anchor has color 0. They occupy distinct replica positions, none equal to the anchor, according to the canceled-anchor extension rule of Lemma 3.3. A constraint $\mathcal{C}$ prescribes any consistent collection of their pairwise split depths, involving only the added colors and the anchor. It imposes no extra restrictions involving the other old replicas. Let $\mathfrak{E}(\mathcal{T}, a_*, \mathcal{C})$ be the allowed extension histories, with coefficients c_E from that lemma. For a history E , set

$$R_E = \frac{1}{N} \sum_{i=1}^N \begin{cases} \sigma_i^{a*} \prod_{j=1}^k \sigma_i^{a_j(E)}, & \text{spin anchor,} \\ \prod_{j=1}^k \sigma_i^{a_j(E)}, & \text{marker anchor.} \end{cases}$$

For $k = 0$ these are respectively $R_{\{a*\}}$ and 1.

Theorem 6.5 (Identity on prescribed trees). *For the perturbation family defined above, along every sequence from Lemma 6.2,*

$$\sum_{E \in \mathfrak{E}(\mathcal{T}, a_*, \mathcal{C})} c_E \mathbb{E}[f R_E] - \mathbb{E} f \sum_{E \in \mathfrak{E}(\{a_*\}, a_*, \mathcal{C})} c_E \mathbb{E}[R_E] \longrightarrow 0. \quad (6.17)$$

This holds for every fixed old tree, either kind of anchor, every bounded polynomial f in its multioverlaps, every finite number of ordered added colors, and every prescribed constraint $\mathcal{C}$. The empty addition is allowed. The error is taken to zero at fixed L , fixed tree, and fixed constraint, before any limit in L .

Proof. For $k = 0$, the claim is the vanishing constant coefficient from Lemma 6.4, using constant marks. For $k \geq 1$, we explain how to extract distinct colored marks. Fix rational-valued functions $g_0, g_1, \dots, g_k$, all bounded by one and constructed jointly from the same finite collection of auxiliary marks. The mixed insertion uses $A = 1 + \sum_{j=1}^k z_j g_j \sigma_i$. Its mixed derivative in all z_j is the symmetric multilinear form obtained by polarizing the degree- k coefficient for a single g . More explicitly, if $Q_N(g) = k! [t^k] H_N(t, 0)$, then the mixed derivative is

$$\frac{k^k}{2^k k!} \sum_{\epsilon \in \{-1, 1\}^k} \left(\prod_{j=1}^k \epsilon_j \right) Q_N \left(k^{-1} \sum_{j=1}^k \epsilon_j g_j \right).$$

Every argument of Q_N is rational-valued and bounded by one, so each resulting pair (g, g_0) belongs to the fixed mark family. Lemma 6.4 makes the displayed mixed derivative tend to zero. The mixed derivative expansion is precisely the ordered extension rule: there is no additional factorial in its coefficients.

It remains to arrange that the expectation over the marks selects exactly the extensions satisfying $\mathcal{C}$. For a pair of colors $a, b \in \{0, \dots, k\}$ and a depth $1 \leq d \leq L - 1$, the event that their split depth is at least d is the event that they share their vertex at depth d . Introduce a new independent Rademacher coordinate at that depth and give both g_a and g_b its factor. Its contribution to their joint mark expectation is 1 if the vertex is shared and 0 otherwise. For a finite collection of such requirements, use different independent coordinates for the different requirements and define each g_a as the product of its assigned factors. Then

$$\mathbb{E}_{\text{marks}} \left[g_0^{a_*} \prod_{j=1}^k g_j^{a_j(E)} \right] = \prod_{(a, b, d) \text{ required}} \mathbf{1}_{\{a, b \text{ share their depth-}d \text{ vertex}\}}. \quad (6.18)$$

All marks in this calculation are independent of the base reservoir.

Let $H_d(a, b)$ denote the sharing indicator in (6.18), with $H_0 = 1$ and $H_L = 0$ for distinct replica positions. These are positions in the sampled tree, even when their spin configurations happen to agree. Exact splitting at depth d is expressed by $H_d - H_{d+1}$, for $0 \leq d \leq L - 1$. Products and finite linear combinations therefore select any prescribed consistent colored shape or collection of allowed shapes. For each term of this finite linear combination, the functions $g_0, \dots, g_k$ are products of signs and hence take values in $\{-1, 1\}$. These functions and their polarization averages belong to the mark family fixed at the start of the section. Constant marks cover $k = 0$.

Finally apply the mixed canceled-anchor expansion to the difference (6.15), average the marks using the selector, and average the independent uniform insertion site. The latter average is exactly R_E . Lemma 6.4 and polarization make the coefficient difference tend to zero, yielding (6.17). Finite linear combinations extend the conclusion from overlap monomials to the stated tests. The construction uses only marks through depth $L - 1$: the extension rule never reuses the anchor or an already used color, so no test of coincidence at depth L is required. $\square$

Remark 6.6 (Order of the choices). The mark specifications and their positive weights c_ν are fixed first, for a fixed L . Parameter averaging then gives one selected sequence on which all types concentrate and all finite overlap moments converge. The analytic argument takes $N \rightarrow \infty$ with each j fixed before sending $j \rightarrow \infty$. Only after the resulting prescribed-tree identities have been established is L allowed to increase. No concentration estimate uniform over the countable family is needed.

7 Decorrelation on prescribed trees

Throughout this section $m_d = d/L$ and $\delta = L^{-1}$. For each fixed L , all limits in N are along the sizes and parameter choices supplied by [Lemma 6.2](#); they satisfy the identities of [Theorem 6.5](#). Different values of L may use different sequences. The argument uses only these identities and the tree sampling rules; in particular, it does not use the interaction factorization.

Our goal is concentration of every finite multioverlap, averaged over separated interior branching depths. These are the averages needed for the comparison with independent messages in [Section 8](#).

7.1 Averages over depths

A topology is a finite rooted tree on labeled leaves, with unary vertices suppressed. Each branching vertex is assigned an integer depth in $\{0, \dots, L-1\}$, increasing strictly along every branch. Write $b(S)$ for the number of branching vertices of a topology S . Fix $\eta > 0$. A depth assignment is η -regular if every assigned depth divided by L lies in $(\eta, 1-\eta)$ and the normalized depths of distinct branching vertices differ by more than η . When an additional prefix depth d is specified, it must satisfy the same restrictions relative to all branching depths. When two copies of one topology use aligned depths, corresponding vertices share one depth coordinate; regularity is imposed on these distinct coordinates.

For a nonnegative array $z(\mathbf{q})$ on a depth coordinates, put

$$\mathcal{A}_{L,\eta} z = L^{-a} \sum_{\mathbf{q} \text{ } \eta\text{-regular}} z(\mathbf{q}). \quad (7.1)$$

Additional ordering restrictions, such as $d < \min_v q_v$, are included in the summation domain. These are unnormalized averages: their total mass is at most one. Consequently, for $0 < \gamma \leq 1$,

$$\mathcal{A}_{L,\eta}(z^\gamma) \leq (\mathcal{A}_{L,\eta} z)^\gamma. \quad (7.2)$$

Two elementary comparisons will recur. Summing over an unused coordinate costs at most L , canceled by its normalization. A fixed integer shift of any subset of the coordinates maps η -regular assignments injectively into $\eta/2$ -regular assignments for all sufficiently large L .

There are only finitely many labeled topologies of any bounded size and, at fixed L , only finitely many assignments of their depths. Thus errors in [Theorem 6.5](#) can be maximized over any of the finite collections used below before taking $N \rightarrow \infty$. We denote such a maximum by $\varepsilon_{N,L,k}$; for each fixed L, k ,

$$\varepsilon_{N,L,k} \longrightarrow 0. \quad (7.3)$$

Constants multiplying this error may depend arbitrarily on L .

Recall that R_S is the multioverlap of the sampled leaves of S , defined in [\(3.1\)](#). The result to be proved is the following.

Theorem 7.1. *For every fixed labeled topology S ,*

$$\lim_{L \rightarrow \infty} \limsup_{N \rightarrow \infty} \mathcal{A}_{L,\eta} \text{Var}(R_S) = 0 \quad (\eta > 0), \quad (7.4)$$

where only the $b(S)$ branching depths are averaged. For a singleton, whose average has no coordinates, $\text{Var}(R_S) \rightarrow 0$ as $N \rightarrow \infty$ at every fixed L .

The proof has two parts. First we control the randomness below a sampled prefix through the squared conditional covariance of the spin-product vector. This is an induction on the number of leaves; delaying the target splits by one level will control its signed extension coefficients. We then use a marker anchor to concentrate the remaining conditional mean at each fixed depth. Averaging over depths is essential when combining the two estimates.

7.2 Conditional covariance and continuity in depth

For a sampled copy of S , set

$$A_S = \left(\prod_{a \in S} \sigma_i^a \right)_{i \leq N}, \quad R_S = \mathbf{1} \cdot A_S, \quad x \cdot y = N^{-1} \sum_{i=1}^N x_i y_i, \quad \|x\|_N^2 = x \cdot x.$$

At a depth d on its initial stem, including its first branching depth, let $\mathcal{F}_d$ be generated by the root and the sampled prefix through d , and define

$$\begin{aligned} \mu_{d,S} &= \mathbb{E}[A_S \mid \mathcal{F}_d], \\ v_N(S, d) &= \frac{1}{N^2} \mathbb{E} \sum_{i,j=1}^N \text{Cov}(A_{S,i}, A_{S,j} \mid \mathcal{F}_d)^2. \end{aligned} \quad (7.5)$$

The expectation includes the root. Every coordinate of A_S and $\mu_{d,S}$ has absolute value at most one, and $0 \leq v_N(S, d) \leq 1$. The next lemma explains why this squared covariance is the useful quantity to induct on.

Lemma 7.2 (Contractions and three copies). *If a random vector $B \in [-1, 1]^N$ is conditionally independent of A_S given $\mathcal{F}_d$, then*

$$\|(A_S - \mu_{d,S}) \cdot B\|_{L^2} \leq v_N(S, d)^{1/4}. \quad (7.6)$$

If X, Y, Z are conditionally independent copies of a bounded random vector given a sigma field $\mathcal{G}$, and C is its conditional covariance matrix, then

$$\mathbb{E}(X \cdot Y - X \cdot Z)^2 \geq \frac{2}{N^2} \mathbb{E} \sum_{i,j} C_{ij}^2. \quad (7.7)$$

Proof. Conditional on $\mathcal{F}_d$ and B , the squared contraction has expectation $N^{-2} \sum_{i,j} C_{ij} B_i B_j$. Its absolute value is at most $N^{-2} \sum_{i,j} |C_{ij}|$. Cauchy–Schwarz, first over i, j and then over the prefix, proves (7.6). For the second assertion, write $\mu = \mathbb{E}[X \mid \mathcal{G}]$. Conditional independence gives

$$\mathbb{E}[(X \cdot Y - X \cdot Z)^2 \mid \mathcal{G}] = \frac{2}{N^2} \text{tr}((C + \mu\mu^\top)C) \geq \frac{2}{N^2} \text{tr}(C^2),$$

since C is positive semidefinite. □

Lemma 7.3 (Averaged continuity in depth). *Fix a topology S and a subset U of its branching vertices. For each η -regular depth assignment, let S^U have the same topology with the depths of the vertices in U increased by one. At an ancestral evaluation depth d , compare the means using the same sampled prefix. Then, for all sufficiently large L ,*

$$\mathcal{A}_{L,\eta} \mathbb{E} \|\mu_{d,S^U} - \mu_{d,S}\|_N^2 \leq C/L. \quad (7.8)$$

The average ranges over all branching depths; the coordinate d is also averaged if it ranges, and otherwise is held fixed. If d ranges along an initial unary stretch of S , then

$$\mathcal{A}_{L,\eta} \mathbb{E} \|\mu_{d+1,S} - \mu_{d,S}\|_N^2 \leq C/L, \quad (7.9)$$

where this average includes d and all branching depths, and the means are evaluated along the same sampled path. The constant depends only on the number of vertices and shifts, uniformly in N , the evaluation depth, and the transition kernels.

Proof. Keep the later branching depths of T fixed and sample a single path along its initial stem. On this path, $\mu_{t,T}$ is a bounded vector martingale. Orthogonality of its increments gives

$$\sum_t \mathbb{E} \|\mu_{t+1,T} - \mu_{t,T}\|_N^2 \leq 1. \quad (7.10)$$

This proves the evaluation-depth assertion after summing over the ranged depth and dividing by L .

Consider a branching vertex at depth q with child shapes $T_1, \dots, T_s$. Holding their later depths fixed, its original conditional mean is $\prod_j \mu_{q,T_j}$, with products understood coordinatewise. If the split is delayed to $q+1$, its conditional mean at the same prefix of depth q is $\mathbb{E}[\prod_j \mu_{q+1,T_j} \mid \mathcal{F}_q]$, where the child means are evaluated on a common one-step continuation. Jensen's inequality and the product difference inequality on $[-1, 1]^s$ yield

$$\mathbb{E} \left\| \prod_j \mu_{q,T_j} - \mathbb{E} \left[\prod_j \mu_{q+1,T_j} \mid \mathcal{F}_q \right] \right\|_N^2 \leq s \sum_j \mathbb{E} \|\mu_{q+1,T_j} - \mu_{q,T_j}\|_N^2. \quad (7.11)$$

Summing over q uses (7.10) with each child shape fixed. Propagating a changed subtree mean towards an ancestral evaluation consists of conditional expectations and multiplication by factors bounded by one. Both operations contract this squared norm. Averaging over all other depth coordinates therefore proves the claimed C/L bound for one shift. For several shifts, change the vertices one at a time and use the squared triangle inequality with a factor equal to the number of shifts. Regularity ensures that the intermediate shapes have the same ancestry relations and, for large L , lie in the enlarged $\eta/2$ -regular domain. All relevant one-prefix marginal laws are unchanged when other sampled branches are omitted, so these estimates hold in any larger sampled tree. $\square$

Suppose now that S first branches at depth e into proper child shapes $T_1, \dots, T_s$. Conditional independence of the children gives $\mu_{e,S} = \prod_j \mu_{e,T_j}$. Write

$$a_{S,e} = \prod_{j=1}^s \mu_{e,T_j}, \quad p_N(S, e) = \sum_{j=1}^s v_N(T_j, e)^{1/4}. \quad (7.12)$$

For any vector $B \in [-1, 1]^N$ independent of the continuation of S given its prefix at e , successive replacement of the children, using (7.6), gives

$$\|(A_S - a_{S,e}) \cdot B\|_{L^2} \leq p_N(S, e). \quad (7.13)$$

At each replacement the other child vectors and the already replaced means form a bounded multiplier independent of the child being replaced, conditionally on $\mathcal{F}_e$. In particular this applies to $B = \mathbf{1}$, and to a vector sampled on another side separated before e . For two such sides, both can be projected with total L^2 error at most $2p_N(S, e)$. Multiplication of the contracted error by any scalar test f with $|f| \leq 1$, even a correlated one, does not increase its L^1 bound.

7.3 Conditional decorrelation

Proposition 7.4. *For every fixed labeled topology S and every $\eta > 0$,*

$$\lim_{L \rightarrow \infty} \limsup_{N \rightarrow \infty} \mathcal{A}_{L,\eta} v_N(S, d) = 0, \quad (7.14)$$

where the average ranges over all branching depths of S and an additional regular depth d strictly before its first split. For a singleton, the average ranges over d alone.

Proof. We induct on the number k of leaves. The induction replaces each proper child by its conditional mean. It then applies the prescribed-tree identity to two copies with delayed branching depths; after summing the signed coefficients, this becomes a nonnegative three-copy estimate. To keep track of the depth averages, let

$$V_{k,L}(\eta) = \max_{1 \leq |S| \leq k} \limsup_{N \rightarrow \infty} \mathcal{A}_{L,\eta} v_N(S, d). \quad (7.15)$$

The maximum is over finitely many labeled topologies. Averages of any lower-order covariance used below, including averages with extra unused coordinates or shifted depths, are bounded by the appropriate $V_{k-1,L}(\eta/2)$, using (7.2) when fractional powers occur.

One leaf. Take two old paths splitting at d , with path 1 a spin anchor, and add one color required to split from the anchor exactly at d . The available old path 2 has coefficient 1. A fresh path can split from both old paths at d , with coefficient $-c_d$, or follow path 2 and leave it at $\ell > d$, with coefficient $-c_\ell$, where

$$c_d = 2m_{d+1} - m_d = (d+2)\delta, \quad c_\ell = \delta \quad (d < \ell \leq L-1). \quad (7.16)$$

From the anchor alone the total coefficient is $-\delta$. Write $X = R_{12}$ and $X_\ell = R_{13}$ for the corresponding fresh extension. Each X_ℓ has the marginal law of X , and $\sum_{\ell=d}^{L-1} c_\ell = 1 + \delta$. The prescribed-tree identity with $f = X$ therefore gives

$$\frac{1}{2} \sum_{\ell=d}^{L-1} c_\ell \mathbb{E}(X - X_\ell)^2 = \delta \text{Var}(X) + o_N(1) \leq \delta + o_N(1). \quad (7.17)$$

For $\ell = d$, the three spin vectors are conditionally independent below d . Hence (7.7) and $c_d \geq \eta$ give $v_N(S, d) \leq \delta/\eta + o_N(1)$ at each fixed L, d . This proves $V_{1,L}(\eta) \leq \delta/\eta$.

Shifted copies and their coefficients. Let $|S| = k \geq 2$, its first split be $e > d$, and its proper child shapes be $T_1, \dots, T_s$. Sample an old tree consisting of two copies of S , denoted I and II, whose stems split at d and whose internal depth assignments are aligned. Fix an old leaf in I as the spin anchor. Let S^+ be the topology S with every internal branching depth increased by one. In the prescribed-tree identity require the anchor and the added colors to form two copies of S^+ , separated at d , with the anchor in its original labeled position in the first copy. These constraints involve only the anchor and the added colors. Add all other colors of side I first, then those of side II. Figure 1 illustrates one local shift.

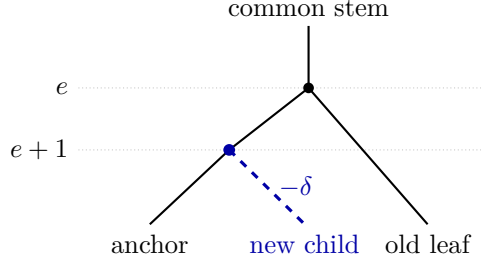


Figure 1: One local effect of delaying a target split. The old tree is solid; the dashed child creates a split at the previously unary depth- $e+1$ vertex, with coefficient $m_{e+1} - m_{e+2} = -\delta$. The proof shifts every internal target depth and charges distinct vertices separately, even when their depths coincide. Later target colors may reuse eligible old leaves.

Let $\kappa(T)$ be the reference coefficient (3.13), and put

$$J = \kappa(S^+)^2. \quad (7.18)$$

At each vertex the $j = 1$ factor is $-\delta$; for $j \geq 2$ its absolute value is at least $\eta/2$ for all sufficiently large L . If $b = b(S)$, then

$$|J| \geq c_{k,\eta} \delta^{2b} > 0. \quad (7.19)$$

For large L , regularity ensures that each shifted depth differs from every branching depth of the old tree, including d . Thus these $2b$ target target branching vertices must be created by fresh unary splits. Lemma 3.5 gives

$$\frac{1}{|J|} \sum_{\text{allowed extensions}} |c| \leq C_{k,\eta}. \quad (7.20)$$

The two copies have aligned depths but distinct branching vertices: each vertex supplies its own factor δ . This is the reason for the shift. The full signed coefficient has an additional factor $-\delta$ from separation at d ; we divide by J and leave that factor in the identity. Equation (7.20) then controls every projection error after division.

Projecting the two target sides. In every extension, the marginal target tree consists of two ordinary copies of S^+ , independent below their separation at d . By (7.13) its multioverlap differs in L^2 from the contraction of its projected means at $e+1$ by at most $2p_N(S^+, e+1)$. Change the proper child depths back by one, then change their evaluation depth from $e+1$ to e . The product difference inequality and Lemma 7.3 show that this adds an error at most $h_N(S)$ in L^2 , with

$$\mathcal{A}_{L,\eta} h_N(S)^2 \leq C_k/L. \quad (7.21)$$

One can take $h_N(S)$ to be the sum of the finitely many $L^2(\|\cdot\|_N)$ distances in these comparisons, multiplied by a size-dependent constant. This definition depends only on the single-copy marginal shapes, not on the particular extension. Thus (7.21) holds uniformly for the comparisons in the extension sum, even though the union of the old and target trees itself need not have regular depths.

Let a, b denote the vectors $a_{S,e}$ on the old stems I and II. The target stem in I shares the old prefix through e because it contains the anchor, so its final projected vector is a .

The signed extension sum. Set f to be the overlap of all $2k$ old leaves, an allowed fixed overlap monomial. All target-I additions stay in the anchor's branch below d , so they leave the old-II stem and its child counts unchanged. Classify the first added path of target side II by its prefix through e . It may diverge freshly at $\ell = d$, with coefficient $-c_d$, or follow the old II stem and diverge at $d < \ell < e$, with coefficient $-c_\ell = -\delta$. In such a class let b_ℓ be $a_{S,e}$ on the new stem; it shares the old II prefix through ℓ . At $\ell = d$ the stems carrying a, b, b_d are conditionally independent below d . Every remaining possibility follows the old II prefix through e and has final projected vector b . All later target paths in side II follow its first path through e , since their earliest split is $e + 1$.

For a fixed extension history, retain every old path and the first target-II prefix through e , and delete all other new branches. The target-I projection a already depends only on the retained old anchor prefix. Each deleted subtree integrates to one under its probability kernels; a reused old leaf introduces no new variable. The retained tree depends only on the class: it is the old tree with a fresh stem attached at $\ell < e$, or just the old tree in the remaining class. The history is a combinatorial choice, not a condition on sampled values. Thus the joint marginal of the old paths and projected prefixes is fixed within each class. In particular, the expectation of f times the projected overlap is constant within a class and can be multiplied by its total signed coefficient.

By Lemma 3.4, summing later side-II colors backwards gives $\kappa(S^+)$ for each first-path choice. Summing the side-I choices gives the other factor $\kappa(S^+)$. The fresh class at ℓ therefore has total coefficient $-c_\ell J$. The total coefficient is $-\delta J$, both from the old tree and from the anchor alone. Since

$$\sum_{\ell=d}^{e-1} c_\ell = (e+1)\delta = m_e + \delta,$$

the remaining prefix-through- e class has coefficient $m_e J$. Thus the projected terms have the following coefficients:

side-II prefix	projected overlap	total coefficient
fresh divergence at $d \leq \ell < e$	$a \cdot b_\ell$	$-c_\ell J$
old prefix through e	$a \cdot b$	$m_e J$

Multiply the projected overlap in each row by f and divide the prescribed-tree identity by J . Rearranging gives

$$\sum_{\ell=d}^{e-1} c_\ell \mathbb{E}[f(a \cdot b - a \cdot b_\ell)] = \delta(\mathbb{E}[f(a \cdot b)] - \mathbb{E}f \mathbb{E}R_{\text{target}}) + \rho_N(S, d), \quad (7.22)$$

where R_{target} on the right is the target overlap in the anchor-alone extension. Its absolute value is at most one. The error satisfies

$$|\rho_N(S, d)| \leq C_{k,\eta}(p_N(S^+, e+1) + h_N(S)) + C_{L,k,\eta}\varepsilon_{N,L,k}. \quad (7.23)$$

Indeed, multiplying a projection error by f , even when correlated with it, costs at most its L^2 norm since $|f| \leq 1$. Sum these errors with absolute coefficients and apply (7.20). The final term includes division of the original identity error by J and tends to zero at fixed L .

Closing the induction. Equation (7.13) gives $\|f - a \cdot b\|_{L^2} \leq 2p_N(S, e)$. Moreover $\sum_{\ell=d}^{e-1} c_\ell = (e+1)\delta \leq 1$. Replacing f by $a \cdot b$ on the left of (7.22) therefore costs at most $4p_N(S, e)$. Each $a \cdot b_\ell$ has the same marginal law as $a \cdot b$, so that left side becomes

$$\frac{1}{2} \sum_{\ell=d}^{e-1} c_\ell \mathbb{E}(a \cdot b - a \cdot b_\ell)^2.$$

Its summands are nonnegative and $c_d \geq \eta$. The right side of (7.22) is bounded in absolute value by $2\delta + |\rho_N(S, d)|$. Hence

$$\mathbb{E}(a \cdot b - a \cdot b_d)^2 \leq C_\eta(\delta + p_N(S, e) + |\rho_N(S, d)|). \quad (7.24)$$

Sample three copies of A_S conditionally independently below d . Projecting each of the two contractions involving the first copy changes it in L^2 by at most $2p_N(S, e)$. Using (7.7), (7.24), and $p_N(S, e) \leq k$, we obtain

$$v_N(S, d) \leq C_{k,\eta}(\delta + p_N(S, e) + p_N(S^+, e+1) + h_N(S)) + C_{L,k,\eta} \varepsilon_{N,L,k}. \quad (7.25)$$

Each child in either p_N has fewer than k leaves. To make the averaging step explicit, the aligned construction has $D = b(S) + 1$ distinct depth coordinates, including d and e . A term belonging to T_j uses only $D_j = b(T_j) + 1$ coordinates. For the shifted term these are $\pi_j(\mathbf{q}) = (e+1, (q_v + 1)_{v \in \text{Br}(T_j)})$. Every retained assignment has at most L^{D-D_j} preimages, and its coordinates are $\eta/2$ -regular for large L . If w_j is the child covariance as a function of these coordinates, then, for $0 < \gamma \leq 1$,

$$\begin{aligned} L^{-D} \sum_{\mathbf{q} \text{ } \eta\text{-regular}} w_j(\pi_j(\mathbf{q}))^\gamma &\leq L^{-D_j} \sum_{\mathbf{y} \text{ } \eta/2\text{-regular}} w_j(\mathbf{y})^\gamma \\ &\leq \left(L^{-D_j} \sum_{\mathbf{y} \text{ } \eta/2\text{-regular}} w_j(\mathbf{y}) \right)^\gamma. \end{aligned} \quad (7.26)$$

The unshifted term has the same bound without translating the coordinates. Thus the two aligned copies use single-copy estimates; there is no restriction of an independently averaged pair of copies to a diagonal. Taking $\gamma = 1/4$ and then $\limsup_N$, and using (7.21) and (7.3), gives, for sufficiently large L ,

$$V_{k,L}(\eta) \leq C_{k,\eta} \left(L^{-1/2} + V_{k-1,L}(\eta/2)^{1/4} \right). \quad (7.27)$$

The maximum over smaller topologies in (7.15) can be absorbed in the right side because these covariances are bounded by one. The base case and induction now imply $V_{k,L}(\eta) \rightarrow 0$ for every fixed k, η , proving the proposition. $\square$

7.4 Concentration of every multioverlap

The covariance estimate controls fluctuations below a sampled prefix. We now use a marker anchor to concentrate the prefix projection, and combine this with the averaged continuity estimate.

Proof of Theorem 7.1. For a singleton, use the spin-anchor identity with no added colors and $f = R_S$. It gives $\mathbb{E}R_S^2 = (\mathbb{E}R_S)^2 + o_N(1)$.

Suppose S first splits at e and has proper child shapes $T_1, \dots, T_s$. We will condition one step before e . Use S as the old tree, designate one of its leaves as a marker anchor, and add a full copy of S all of whose leaves split from the anchor at $e-1$. Its first path must be fresh at $e-1$, because the old tree has not yet branched there. All other target paths lie in the resulting fresh subtree. The extension coefficient on both sides of the prescribed-tree identity is the same nonzero number

$$C(S, e) = -\delta\kappa(S). \quad (7.28)$$

The target and old copies are conditionally independent and identically distributed below $\mathcal{F}_{e-1}$. With $f = R_S$ and $M = \mathbb{E}[R_S \mid \mathcal{F}_{e-1}]$, their product has conditional expectation M^2 . On the anchor-alone side, deleting the marker's continuation below $e-1$ integrates a probability kernel to one and leaves the ordinary target law S , with mean $\mathbb{E}R_S$. The identity therefore gives

$$C(S, e)(\mathbb{E}M^2 - (\mathbb{E}R_S)^2) = C(S, e) \text{Var}(M) = o_N(1), \quad (7.29)$$

and hence

$$\text{Var}(M) \longrightarrow 0 \quad \text{at each fixed } L, S, e. \quad (7.30)$$

Division by $C(S, e)$ is made only at fixed L ; no uniform lower bound as $L \rightarrow \infty$ is needed.

It remains to control the conditional variance below this prefix. The child covariances are evaluated at e , which is separated from all their branching depths; the move to $e-1$ will use martingale continuity instead. Set

$$G = \mathbf{1} \cdot \prod_j \mu_{e, T_j}, \quad G_- = \mathbf{1} \cdot \prod_j \mu_{e-1, T_j}.$$

The latter is $\mathcal{F}_{e-1}$ -measurable. By (7.13), $\mathbb{E}(R_S - G)^2 \leq p_N(S, e)^2$. Keeping all proper child depths fixed, the product difference inequality and (7.10) imply

$$\mathcal{A}_{L, \eta} \mathbb{E}(G - G_-)^2 \leq C_k/L. \quad (7.31)$$

Here e is itself a ranged coordinate, so the one-step increments are summed before division by L . Since conditional expectation is the optimal L^2 projection,

$$\mathbb{E}(R_S - M)^2 \leq \mathbb{E}(R_S - G_-)^2 \leq 2p_N(S, e)^2 + 2\mathbb{E}(G - G_-)^2.$$

Variance decomposition and (7.30) now show

$$\limsup_{N \rightarrow \infty} \mathcal{A}_{L, \eta} \text{Var}(R_S) \leq C_k \left(L^{-1} + V_{k-1, L}(\eta/2)^{1/2} \right) \quad \text{for all sufficiently large } L. \quad (7.32)$$

In this estimate we used $p_N(S, e)^2 \leq s \sum_j v_N(T_j, e)^{1/2}$ and (7.26) with $\gamma = 1/2$: here the full average has $D = b(S)$ coordinates because d is absent, while still $D_j = b(T_j) + 1$. Proposition 7.4 finishes the proof. $\square$

Remark 7.5. All uncanceled small denominators in this section occur only in prescribed-tree identity errors. At fixed L there are finitely many relevant trees, so those errors disappear before any depth average or L limit is taken. Errors that survive the N limit are controlled by the uniform relative absolute-cost bound (7.20), the lower-order covariance averages, and the martingale estimate C/L . No rate of convergence in N uniform in L is used.

8 Independent hierarchical messages and the cavity comparison

We turn the reservoir into a trial law in three steps: encode its full one-site branching law, use multioverlap concentration to separate the sites, and compare the resulting insertion logarithms.

Fix $\varepsilon > 0$. At each depth $L \geq 2$, use the sequence of sizes and deterministic parameter choices supplied by Lemma 6.2. Write N for sizes on this sequence. The choices may depend on L , and all statements below concern these choices. Retain $\underline{F}$ for the liminf over *all* system sizes of the bounded model, as before selection. In particular,

$$\Delta_N^{L, \mathbf{u}_{L,N}} \leq \underline{F} + 2\varepsilon. \quad (8.1)$$

8.1 Encoding a single-site tree

Lemma 8.1 (Recursive law encoding). *For each fixed N , $L \geq 2$, and selected parameter vector, with the reservoir root disorder averaged, there is a deterministic trial law $\zeta_{L,N} \in \mathcal{M}_L$ whose single-label spin process has the same expected spin products on every prescribed tree as a uniformly chosen site of the reservoir. Different trial labels can be chosen independently.*

Proof. Include an independent uniform site $i \in [N]$ in the root variable, along with the reservoir root disorder. Sample the tilted reservoir auxiliary kernels along each path, and let $\mathcal{G}_d$ be the sigma-field generated by this enlarged root and the full auxiliary prefix through depth d . At depth $r = L - 1$ put

$$x = \operatorname{arctanh} \langle \sigma_i \rangle,$$

where the brackets denote the leaf Gibbs expectation. This is finite: at finite N all configuration weights are strictly positive, so $-1 < \langle \sigma_i \rangle < 1$. Thus x is determined at depth r , before the terminal spin transition at depth L . At each terminal replica leaf, draw a spin with law q_x . Given the auxiliary prefix, these spins are independent and have exactly the one-site law of the independent terminal Gibbs replicas.

Define the random measures backward by

$$\eta_{r-1} = \operatorname{Law}(x \mid \mathcal{G}_{r-1}), \quad \eta_d = \operatorname{Law}(\eta_{d+1} \mid \mathcal{G}_d) \quad (0 \leq d < r - 1).$$

Then $\eta_d \in \mathcal{M}_{r-d}$, so $\zeta_{L,N} = \operatorname{Law}(\eta_0) \in \mathcal{M}_L$. All state spaces are standard Borel, so these conditional probability kernels have measurable versions. The auxiliary priors used here may be taken to be finite products of finite mark spaces at each realized root; hence the same assertion also follows directly by finite sums conditional on the root counts.

Here is an explicit verification for every branching pattern. Fix a descendant tree T and prescribe a spin e_a at each terminal leaf. At depth r , its conditional probability is $G_T(x) = \prod_a q_x(e_a)$. At depth $d < r$, let $T_1, \dots, T_j$ be the subtrees rooted at its children. Conditional independence of children gives the backward recursion

$$G_T(\eta_d) = \prod_{\ell=1}^j \int G_{T_\ell}(\nu) \eta_d(d\nu).$$

By induction, this probability depends on the full prefix only through η_d . Thus the nested measures preserve the joint output law of every descendant tree, including unary stretches and arbitrary numbers of children. This recursion is precisely the branching version of (2.7). Taking $\zeta_{L,N} = \operatorname{Law}(\eta_0)$ preserves the root mixture as well. In particular, for every subset S of the leaves,

$$\mathbb{E} \prod_{a \in S} \tau^a = \frac{1}{N} \sum_{i=1}^N \mathbb{E} \prod_{a \in S} \sigma_i^a = \mathbb{E} R_S. \quad (8.2)$$

Use independent copies of the entire construction, including the reservoir root and uniform site, for different labels. The resulting messages are independent of the new physical disorder in the cavity insertions. $\square$

8.2 From multioverlaps to spin patterns

Lemma 8.2 (Empirical-pattern comparison). *Fix a prescribed tree with k leaves. Suppose an observable of absolute value at most one uses the replica-spin patterns at a independent uniform site labels. Replacing these patterns by a independent trial-label patterns from [Lemma 8.1](#) changes its expected value by at most*

$$a \sum_{\emptyset \neq S \subseteq [k]} \mathbb{E}|R_S - \mathbb{E}R_S|. \quad (8.3)$$

The observable may also depend on new insertion disorder, independent of the reservoir.

Proof. Conditional on the sampled reservoir configurations, the pattern $\sigma_i = (\sigma_i^1, \dots, \sigma_i^k)$ at a uniform site has empirical distribution

$$\pi(e) = \frac{1}{N} \sum_i \prod_{b=1}^k \frac{1 + e_b \sigma_i^b}{2} = 2^{-k} \sum_{S \subseteq [k]} R_S \prod_{b \in S} e_b, \quad e \in \{-1, 1\}^k.$$

The trial-label distribution is $\bar{\pi} = \mathbb{E}\pi$ by (8.2). Independent uniform sampling with replacement gives $\pi^{\otimes a}$ conditionally in the reservoir; the trial distribution is $\bar{\pi}^{\otimes a}$. The elementary telescoping bound for product measures gives

$$\|\pi^{\otimes a} - \bar{\pi}^{\otimes a}\|_{\text{TV}} \leq a \|\pi - \bar{\pi}\|_{\text{TV}} \leq \frac{a}{2} \sum_{\emptyset \neq S \subseteq [k]} |R_S - \mathbb{E}R_S|.$$

Integrating a function bounded by one costs at most twice total variation. Conditioning on the independent insertion disorder proves the last assertion. $\square$

Let Q_k be the tree law in [Lemma 3.2](#). We need the following consequence of the decorrelation theorem:

$$\lim_{L \rightarrow \infty} \limsup_{N \rightarrow \infty} \mathbb{E}_{Q_k} \sum_{\emptyset \neq S \subseteq [k]} \mathbb{E}|R_S - \mathbb{E}R_S| = 0 \quad (k \text{ fixed}). \quad (8.4)$$

Here and below the inner limit is along the selected fixed- L sequence. To see this, fix $0 < \eta < 1/2$, and let $\mathfrak{S}_k$ be the finite collection of labeled topologies on nonempty subsets of $[k]$. We claim the bound

$$\mathbb{E}_{Q_k} \sum_{\emptyset \neq S \subseteq [k]} \mathbb{E}|R_S - \mathbb{E}R_S| \leq C_k \left(\eta + L^{-1} + \sum_{U \in \mathfrak{S}_k} (\mathcal{A}_{L,\eta} \text{Var}(R_U))^{1/2} \right). \quad (8.5)$$

The average for a singleton has no depth coordinates. By [Lemma 3.6](#), nonregular trees have probability at most $C_k(\eta + L^{-1})$. On regular trees every induced subtree retains the interior and separation conditions on its branching depths. For a topology with a branching vertices, each depth assignment has probability at most $C_k L^{-a}$. If an induced subtree retains a' branching vertices, each assignment of its depths has at most $L^{a-a'}$ extensions to the full tree. Summing out these unused coordinates leaves exactly the normalization $L^{-a'}$ in $\mathcal{A}_{L,\eta}$. Cauchy–Schwarz bounds the resulting average of mean absolute deviations by $(\mathcal{A}_{L,\eta} \text{Var}(R_U))^{1/2}$. Summing over the finitely many topologies and subsets, and using the bound two for each deviation on nonregular trees, proves (8.5). Apply [Theorem 7.1](#) first at fixed η , taking the fixed- L limit in N before $L \rightarrow \infty$, and then let $\eta \downarrow 0$ to obtain (8.4).

8.3 Comparison of the insertion logarithms

Proposition 8.3. *For the trial laws of Lemma 8.1, with $\mathbf{m}^{(L)} = (1/L, \dots, (L-1)/L)$,*

$$\lim_{L \rightarrow \infty} \limsup_{N \rightarrow \infty} \left| \Delta_N^{L, \mathbf{u}_{L,N}} - \mathcal{B}_{L-1}(\zeta_{L,N}, \mathbf{m}^{(L)}) \right| = 0. \quad (8.6)$$

Proof. For each insertion in Lemma 5.1, replace the reservoir spins at its site labels by independent spin processes from Lemma 8.1. Denote the resulting trial insertions by $\tilde{A}_s$ and $\tilde{A}_b$, and their recursions from (3.3) by $\tilde{Y}_{0,s}$ and $\tilde{Y}_{0,b}$. These recursions use the trial kernels.

At the terminal spin transition, averaging $\tilde{A}_s$ over its independent trial spins gives exactly V_s in (2.8); hence $\tilde{Y}_{0,s} = T_0 V_s$. For $\tilde{A}_b$ the same transition gives a product of K_b edge factors with disjoint message labels. Given the physical disorder and all message prefixes at a depth, these label groups remain independent. Their product therefore factors through each conditional power mean, successively from the last level to the first. Consequently,

$$\begin{aligned} \mathbb{E} \log \tilde{Y}_{0,b} &= \alpha(p-1) \mathbb{E} \log T_0 V_e, \\ \log 2 + \mathbb{E} \log \tilde{Y}_{0,s} - \mathbb{E} \log \tilde{Y}_{0,b} &= \mathcal{B}_{L-1}(\zeta_{L,N}, \mathbf{m}^{(L)}). \end{aligned}$$

By (5.5), it remains to compare the reservoir and trial insertion logarithms in the stated limits.

Consider either insertion and first restrict its Poisson count K to $K \leq M$, where M is fixed. Set $C = H + BM$, $z = \cosh C$, and $c = \tanh C < 1$. On this event both the reservoir insertion and its trial version can be written as $A = z(1 + D)$ with $|D| \leq c$, uniformly in all disorder and path variables. The deterministic term $\log z$ cancels in the comparison.

At each order k of Lemma 3.2, the product of the k copies of D is a bounded function of replica-spin patterns at at most pM labels and of independent insertion disorder. By Lemma 8.2, its evaluation error on a fixed tree is bounded by $pM \sum_{S \neq \emptyset} \mathbb{E} |R_S - \mathbb{E} R_S|$. Hence (8.4) makes its tree-averaged error tend to zero in the ordered limits. For every fixed M , truncating the expansion after q terms leaves an error at most $\sum_{k>q} c^k/k$ on either side, uniformly in N, L .

Outside the count cutoff the original insertion logarithms, on both sides, are bounded by $H + BK$. Their expected contribution is therefore at most $2\mathbb{E}[(H + BK)\mathbf{1}_{\{K>M\}}]$, which tends to zero as $M \rightarrow \infty$. This proves convergence of the expected insertion logarithms: explicitly, first choose M to control the Poisson tail, then q to control the geometric tail, and then take the limits in N and L for the finitely many remaining expansion terms. Together with the trial identities above and (5.5), this proves (8.6). $\square$

Corollary 8.4. *For every bounded permutation-invariant interaction law and bounded field law, with the functionals defined in Section 2,*

$$\liminf_{N \rightarrow \infty} F_N \geq \inf_{r \geq 0} \Phi_r.$$

Proof. Write $A = \inf_{r \geq 0} \Phi_r$, which is finite by (2.12). Each encoded trial value is at least A , so (8.1) gives, at every fixed L ,

$$A \leq \underline{F} + 2\varepsilon + \limsup_{N \rightarrow \infty} \left| \Delta_N^{L, \mathbf{u}_{L,N}} - \mathcal{B}_{L-1}(\zeta_{L,N}, \mathbf{m}^{(L)}) \right|.$$

The inner limit uses that depth's selected sequence. Letting $L \rightarrow \infty$ in Proposition 8.3 gives $A \leq \underline{F} + 2\varepsilon$, and then $\varepsilon \downarrow 0$ proves the result. $\square$

9 Normalization, truncation, and completion of the proof

The lower bound proved in [Corollary 8.4](#) requires bounded permutation-invariant interactions, whereas the upper bound uses the factorization and positivity hypotheses. An integrable constant shift and uniform approximation connect the two statements.

Lemma 9.1 (Stability). *Couple two interaction laws θ, θ' and two field laws h, h' , each satisfying (2.1) for the fixed p and α , and put $D = \mathbb{E} \max_s |\theta(s) - \theta'(s)|$ and $d = \mathbb{E}|h - h'|$. Then*

$$|F_N - F'_N| \leq \alpha D + d, \quad (9.1)$$

$$|\mathcal{B}_r(\zeta, \mathbf{m}) - \mathcal{B}'_r(\zeta, \mathbf{m})| \leq \alpha(2p-1)D + d \quad (9.2)$$

uniformly in N, r and all trial laws, including $r = 0$. If c is an integrable random constant associated with an interaction and $\hat{\theta} = \theta - c$, then

$$\hat{F}_N = F_N - \alpha \mathbb{E}c, \quad \hat{\mathcal{B}}_r = \mathcal{B}_r - \alpha \mathbb{E}c. \quad (9.3)$$

Proof. Use the same counts and indices in the two Hamiltonians. Their pointwise log-weight difference is bounded by the sum of the interaction sup-norm differences and field differences. The logarithm of the partition function preserves this bound, which gives (9.1). Each cavity interaction changes by at most the corresponding sup-norm interaction difference, and so does each edge logarithm. The power means preserve these root-measurable bounds. The site term has mean αp interactions and the edge term has coefficient $\alpha(p-1)$, giving (9.2).

For (9.3), the physical constants factor out of the spin partition function and of every power mean. In the functional, the site term shifts by $-\alpha p \mathbb{E}c$ and the edge term contributes $+\alpha(p-1)\mathbb{E}c$, for a net shift $-\alpha \mathbb{E}c$. $\square$

Proof of Theorem 2.1. The multiplier a may depend asymmetrically on the labeled input functions, so the original interaction law need not be permutation invariant. We first remove this multiplier by an integrable scalar shift. Set $c = \theta(1, \dots, 1)$ and $\hat{\theta} = \theta - c$. The constant is integrable, since $|c| \leq B(\theta)$. The factorization gives

$$\hat{\theta}(s) = \log \frac{1 + b \prod_{\ell=1}^p f_\ell(s_\ell)}{1 + b \prod_{\ell=1}^p f_\ell(1)}.$$

The multiplier a has disappeared. Independence and identical distribution of the f_ℓ , and their independence of b , show that the law of $\hat{\theta}$ is invariant under permutations of its inputs. Moreover $\mathbb{E}B(\hat{\theta}) \leq 2\mathbb{E}B(\theta) < \infty$.

For $K > 0$, truncate each entry of $\hat{\theta}$ to $[-K, K]$ and truncate h to $[-K, K]$, obtaining $\hat{\theta}^{(K)}, h^{(K)}$. Permutation invariance is preserved. The bounded-law lower bound [Corollary 8.4](#) applies whether or not truncation preserves the factorization. By dominated convergence,

$$D_K = \mathbb{E} \max_s |\hat{\theta}(s) - \hat{\theta}^{(K)}(s)| \longrightarrow 0, \quad d_K = \mathbb{E}|h - h^{(K)}| \longrightarrow 0.$$

Write $\hat{I}$ for the infimum of the normalized trial functional over all depths and trial laws, and I_K for the same infimum with the truncated physical laws. The uniform trial estimate gives

$$|\hat{I} - I_K| \leq \alpha(2p-1)D_K + d_K.$$

The pressure estimate and [Corollary 8.4](#) now imply

$$\liminf_N \hat{F}_N \geq I_K - \alpha D_K - d_K \geq \hat{I} - 2\alpha p D_K - 2d_K.$$

Letting $K \rightarrow \infty$ proves $\liminf_N \widehat{F}_N \geq \widehat{I} = \inf_r \widehat{\Phi}_r$. Undoing the constant shift gives $\liminf_N F_N \geq \inf_r \Phi_r$. Finally Proposition 4.1 gives $\limsup_N F_N \leq \inf_r \Phi_r$. Both limits agree, proving existence and (2.13). $\square$

10 Named models and the zero-temperature limit

The standard examples of the factorization are the diluted spin models and random satisfiability models of Franz–Leone and Panchenko–Talagrand [9, 16]. We retain the Poisson counts and sampling with replacement specified in (2.2).

Corollary 10.1 (Viana–Bray and diluted even-spin models). *Fix an even integer $p \geq 2$, $\alpha > 0$, and independent real random variables J, H such that J has a symmetric distribution and $\mathbb{E}|J| + \mathbb{E}|H| < \infty$. For every $\beta > 0$, the model with*

$$\theta_\beta(s_1, \dots, s_p) = \beta J \prod_{\ell=1}^p s_\ell, \quad h_\beta = \beta H$$

satisfies the variational formula (2.13), with the trial functional evaluated at these interaction and field laws. In particular, this holds for the Viana–Bray model [18], which is the case $p = 2$.

Proof. Use

$$a = \cosh(\beta J), \quad b = \tanh(\beta J), \quad f_\ell(s) = s.$$

The functions f_ℓ are deterministic independent copies, and $|b| < 1$ almost surely. Symmetry makes every odd moment of b zero; the even moments are nonnegative. Finally, $B(\theta_\beta) = \beta|J|$ and $|h_\beta| = \beta|H|$. Thus all hypotheses of Theorem 2.1 hold. $\square$

Corollary 10.2 (Soft random even- K SAT). *Fix an even integer $K \geq 2$ and $\alpha > 0$. Let $\varepsilon_1, \dots, \varepsilon_K$ be independent fair signs and let $W \geq 0$ be independent of the signs, with $\mathbb{E}W < \infty$. For every $\beta > 0$, the interaction*

$$\theta_\beta(s_1, \dots, s_K) = -\beta W \prod_{\ell=1}^K \frac{1 + \varepsilon_\ell s_\ell}{2}, \quad h = 0,$$

satisfies (2.13). The choice $W = 1$ is the soft random even- K SAT model; W gives an independent nonnegative clause weight.

Proof. The product $I_\varepsilon(s) = \prod_\ell (1 + \varepsilon_\ell s_\ell)/2$ takes values in $\{0, 1\}$ and indicates a violated clause. Therefore

$$e^{\theta_\beta(s)} = 1 + (e^{-\beta W} - 1)I_\varepsilon(s).$$

Take $a = 1$, $b = e^{-\beta W} - 1$, and $f_\ell(s) = (1 + \varepsilon_\ell s)/2$. The functions are independent copies and independent of b . Since $0 \leq -b < 1$, the strict product condition and every positivity inequality hold. Moreover, $B(\theta_\beta) = \beta W$, so Theorem 2.1 applies. $\square$

For the spin model, define the unscaled energy score by

$$\mathcal{E}_N(\sigma) = \sum_{k=1}^{M_N} J_k \prod_{\ell=1}^p \sigma_{i_\ell, k} + \sum_{i=1}^N H_i \sigma_i.$$

For the satisfiability model, put

$$\mathcal{E}_N(\sigma) = - \sum_{k=1}^{M_N} W_k \prod_{\ell=1}^K \frac{1 + \varepsilon_{\ell,k} \sigma_{i_{\ell,k}}}{2}.$$

These scores and their disorder laws do not depend on β ; in both cases the partition function is $Z_N(\beta) = \sum_{\sigma} e^{\beta \mathcal{E}_N(\sigma)}$. Let $\mathcal{B}_{r,\beta}$ denote the functional (2.11) for the corresponding laws at inverse temperature β .

Corollary 10.3 (Zero-temperature variational limit). *For each model in Corollaries 10.1 and 10.2, the limit*

$$g = \lim_{N \rightarrow \infty} \frac{1}{N} \mathbb{E} \max_{\sigma} \mathcal{E}_N(\sigma)$$

exists and satisfies

$$g = \lim_{\beta \rightarrow \infty} \frac{1}{\beta} \inf_{r \geq 0} \inf_{\zeta, \mathbf{m}} \mathcal{B}_{r,\beta}(\zeta, \mathbf{m}). \quad (10.1)$$

More precisely, the expression inside the temperature limit lies in $[g, g + \log(2)/\beta]$. For unit-weight SAT, the limiting expected minimum number of violated clauses per variable is $-g$.

Proof. Write $g_N = N^{-1} \mathbb{E} \max_{\sigma} \mathcal{E}_N(\sigma)$ and $P_N(\beta) = N^{-1} \mathbb{E} \log Z_N(\beta)$. The first-moment assumptions make g_N finite and uniformly bounded. Since there are 2^N spin configurations,

$$g_N \leq \frac{P_N(\beta)}{\beta} \leq g_N + \frac{\log 2}{\beta}.$$

For each fixed $\beta > 0$, Theorem 2.1 gives $P_N(\beta) \rightarrow I(\beta)$, where $I(\beta) = \inf_{r,\zeta,\mathbf{m}} \mathcal{B}_{r,\beta}(\zeta, \mathbf{m})$. Hence

$$\limsup_N g_N \leq \frac{I(\beta)}{\beta} \leq \liminf_N g_N + \frac{\log 2}{\beta}.$$

Letting $\beta \rightarrow \infty$ proves that g_N converges to a limit g . Taking $N \rightarrow \infty$ in the finite-volume bounds now gives $g \leq I(\beta)/\beta \leq g + \log(2)/\beta$, proving (10.1). In the satisfiability model, maximizing the negative violation count is the same as minimizing that count. $\square$

The infimum in (10.1) is taken over all finite hierarchies separately at each positive temperature. The formula does not require attainment or a limiting hierarchy.

Remark 10.4. The even-arity hypothesis in Corollary 10.2 excludes random 3-SAT. For that model, the companion [13, Theorem 8.4] gives an exact soft-pressure formula over finite rational trials for a Poisson number of clauses whose variable positions are sampled independently with replacement. Its sign-indexed message pairs need not form a probability distribution, and its trial functional differs from (2.11).

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